Following to Lead: Dominant Followership in Sequential Price Competition

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Abstract

Dominant firms are usually thought to be price leaders: they move first, and rivals follow. Amazon, the dominant platform in online retail, rather follows than leads. Further, despite its dominant market position, Amazon is usually priced at or below the lowest rival price; its distinctiveness lies not in what it charges but in how it responds when a rival moves first. Using synchronized high-frequency cross-platform pricing data, I document three patterns in Amazon's response. First, after a rival raises its price, Amazon is 13 percentage points more likely to raise its own price than it would be otherwise, compared with no detectable response from rivals when Amazon moves first. Second, when Amazon responds, it matches the rival's new price. Third, matched rival price increases remain in place longer than unmatched ones. I call this strategy dominant followership. A sequential Bertrand model modified to break ties in favor of the dominant firm shows why dominant followership can be incentive-compatible: because the dominant firm keeps most demand when prices are comparable, it may prefer matching a rival's higher price to undercutting it. Prices rise above the simultaneous-move Nash equilibrium only when the demand-advantaged firm is also the reliable responder. Using simulation studies, I show that Q-learning algorithms discover dominant followership endogenously: the rival's algorithm learns to initiate higher prices, nearly tripling the market price, yet the dominant firm captures 91 percent of the resulting profit gains. This speaks to the FTC's Project Nessie allegation: individually optimal responses can teach rival algorithms that raising price is profitable, even without explicit coordination. The main insight is that modern algorithmic price competition among asymmetric firms should be modeled as a sequential response process: who responds to a price change matters as much as the price itself.

Keywords: **Algorithmic Pricing, Artificial Intelligence, Sequential Price Competition, Demand Asymmetry, Reinforcement Learning**

JEL Classification: **L13, L41, C63, C73, L81**

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1 Introduction

The rise of AI-driven algorithmic pricing is changing the nature of price competition. Pricing algorithms allow firms to characterize not only optimal price levels but also optimal responses to rivals' price adjustments. These algorithmic price-response functions are becoming increasingly important as AI-driven pricing tools make price responses more scalable and adaptive.[1] They have already attracted regulatory attention: the U.S. Federal Trade Commission (FTC) has accused Amazon's internal pricing algorithm "Project Nessie" of using competitors' pricing responses to guide the entire market toward collusive higher prices (Federal Trade Commission 2024).[2]

The conventional view of price leadership holds that firms with market power set prices first, and rivals follow. Despite its market position, Amazon does the reverse: it is the one that follows. Further, it is rarely the most expensive option, typically pricing at or below its rivals. But that is not where its distinctiveness lies. The distinctive pattern is how it responds to rival price changes: rivals move first, and Amazon follows. This raises a natural question: can a dominant firm raise market prices by following rivals rather than leading them? In this paper, I ask why this pattern arises and what it implies for modeling modern algorithmic price competition.

Answering this requires attention to two features of competition that standard models of simultaneous competition among symmetric firms often ignore. The first is demand advantage: some platforms attract customers through convenience, switching frictions, or other non-price factors, giving them a source of loyalty that has nothing to do with charging less. Amazon is a natural example, allegedly accounting for more than 82 percent of gross merchandise value in the online-superstore market defined in the FTC complaint. The second is response reliability: some platforms respond to rivals' price changes far more systematically than others. These two forces need not coincide. But when a demand-advantaged platform is also the reliable responder, rival price increases are more likely to persist, raising the price the market ultimately pays.

Using a custom scraper that tracks Amazon's first-party prices against every rival platform

[1] Among 419 B2B pricing executives and decision-makers, 65 to 85 percent reported that their organizations expected to adopt generative and agentic AI in pricing within one to three years; see McKinsey, "B2B pricing: Navigating the next phase of the AI revolution," April 7, 2026.

[2] A related California enforcement action alleges coordinated price increases in which one retailer raised its price and another subsequently matched the higher price. Unlike the mechanism studied here, which operates through independently chosen price responses and does not require coordination, the California filing alleges vendor-mediated agreements among Amazon and competing retailers. See *People v. Amazon.com, Inc.*, Memorandum in Support of Motion for Preliminary Injunction (Cal. Super. Ct. 2026), pp. 2–3.

selling the exact same product, every two hours over eight months, I document how Amazon and its rivals respond to each other's price changes in more than one million synchronized price observations. Amazon's response unfolds in three parts, together explaining how the higher price is sustained. First, it reacts far more often than rivals do: after a rival-initiated price increase, Amazon is 12.9 percentage points more likely to raise its own price within 48 hours, 3.5 times the response of other major platforms, who respond by only 3.7 points. And when Amazon itself moves first, other major platforms respond by just 0.15 points. Second, when Amazon does react, it typically matches rather than undercutting the rival's new price: 40.3 percent of its responses land exactly at that price. Third, this matching helps the higher price stick: matched rival price increases are 15.8 percentage points more likely to remain in place after two weeks than unmatched increases, while matched decreases show no such effect. I call this strategy "dominant followership."

The dominant followership documented in the data can be rationalized in a simple Bertrand model with one seemingly minor modification: one firm has a demand advantage at price parity. One way to microfound this assumption is a lexicographic decision rule: consumers choose the lower price, but when prices are equal, they favor the dominant platform. This keeps the model close to homogeneous-products Bertrand competition, while allowing platform advantages to matter. The key insight is that this advantage gives the dominant firm little incentive to undercut the rival's higher price; it may instead prefer to match it, monetizing the demand it already holds. The rival, holding only a minority of that same tied-price demand, has every incentive to undercut instead. But this undercutting incentive pushes prices toward marginal cost only when the rival can react quickly. A market friction that limits the rival's ability to react can therefore sustain equilibrium prices above the simultaneous-move Nash equilibrium. The single-period model captures this friction through leader-follower timing. The infinite-horizon model extends it by giving the rival only occasional response opportunities, while the dominant firm can respond reliably. This extension explains both sides of the resulting pattern. The rival initiates a costly price increase only if it is patient enough to value the dominant firm's eventual match more than the demand it sacrifices in the short run. The resulting higher price then tends to persist when the rival's own opportunities to respond are only occasional.

The model explains why dominant followership can soften competition when firms understand the incentives they face. In real-world online retail, however, these incentives may be difficult to know in advance: retailers set prices across large product assortments where demand conditions and rival pricing patterns differ across products and evolve over time. I therefore model this learning process with Q-learning simulations, in which firms are not told to follow the dominant-followership strategy but must discover it on their own from realized profit feedback. Existing Q-learning models of algorithmic pricing typically study symmetric firms, whether they move simultaneously or alternate sequentially with symmetric response opportunities. I depart from this benchmark in two empirically motivated ways: firms differ in demand advantage, and response reliability is also asymmetric, with the dominant firm responding more reliably[3] than the rival does. The channel operates only when both forces coincide. Compared with a baseline in which neither asymmetry is present, demand asymmetry alone raises the learned market price by only about 12 percent, asymmetric response timing alone by 37 percent, and both combined by nearly 70 percent.

The rival's algorithm does all the work of raising prices; the dominant firm keeps nearly all the reward. Since the theory shows that the dominant firm's willingness to match is what sustains the higher price, one might expect its algorithm, not the rival's, to be the one that must learn to make this mechanism work. To identify whose learning matters, I let one firm at a time use Q-learning while the other follows a simpler, myopic pricing rule. This confirms the reversal: when only the dominant firm learns, the market price remains low; when only the rival learns, it reaches 89 percent of the price level achieved when both firms learn. The reason is a stark asymmetry in what each firm has to figure out. For the dominant firm, matching a higher rival price is a one-shot decision, requiring no experimentation. For the rival, initiating that higher price is a gamble: it only learns whether the bet pays off after repeatedly absorbing the short-run cost of raising price first and waiting to see how the dominant firm responds. Yet once the rival's algorithm learns that initiation can be profitable, the dominant firm captures 91 percent of the resulting profit gains. This has a direct bearing on cases like the Project Nessie allegation: if it is a rival's own algorithm that learns to raise prices, scrutiny aimed only at a dominant platform's pricing algorithm may be looking in the wrong place.

The main insight of this paper is that modern price-response algorithms should be modeled as sequential response processes. This challenges the tacit assumption that simultaneous-move games

[3] I use "reliable responder" to refer to the model counterpart of the empirical follower role: the firm that can respond to rival price changes more consistently than its rival can.

provide an adequate description of long-run retail price competition. I show that the key object in price competition is not only the price a firm sets, but also whose price change becomes observable first and who responds. Simultaneous-move models miss this channel because they remove the responder role through which demand advantage shapes incentives and learning. They therefore fail to approximate long-run outcomes when firms differ in both demand advantage and response reliability.

The paper connects three literatures through one mechanism: dominant followership in sequential price competition among asymmetric firms. First, I provide new evidence on a response margin that standard pricing data usually miss. Using novel high-frequency cross-platform data, I show that Amazon plays the distinctive role of a follower, matching rival price increases in a way that helps those higher prices persist. Second, I introduce dominant followership as a new mechanism through which demand asymmetry can soften competition. The mechanism shifts attention from price leadership to anticipated response: a dominant firm can raise market prices not by moving first, but by responding reliably enough to make raising price profitable for the rival. This overturns the conventional expectation that advantage belongs to the first mover. Third, I show this mechanism can emerge endogenously from algorithmic learning. The learning result separates initiation from incidence: the rival's algorithm generates the price increase, while most of the gains accrue to the dominant firm.

These findings change how algorithmic pricing should be evaluated. Dominant followership need not look anticompetitive in isolation. The dominant firm is not necessarily leading prices upward, communicating with rivals, or committing to a common pricing rule. It may simply be responding to a rival's price as an individually optimal response. I therefore do not label the mechanism as implicit collusion. Each firm is choosing the price response that is privately optimal given its own demand position. The concern is that, once these responses become systematic, they can enter rival algorithms' learning environments and raise market-wide prices. This implies that regulators and modelers should pay attention not only to price levels, but also to response patterns, timing, and which firms' algorithms learn from those responses. For a firm weighing whether to adopt a learning algorithm, the results carry a specific caution: it may be your own algorithm that learns to raise prices, while a dominant rival is the one who profits from it.

Related Literature. This paper contributes to the empirical literature on algorithmic pricing. Existing empirical work primarily studies competitive pricing behavior or whether the adoption or use of pricing algorithms affects price levels or margins in particular markets, including automobiles, gasoline, housing, online marketplaces, and earlier digital-pricing settings (Sudhir 2001; Kannan and Kopalle 2001; Assad, Clark, Ershov, and Xu 2024; Calder-Wang and Kim 2023; Musolff 2024; Zhao and Berman 2025). Less is known about how algorithmic pricing operates through responsiveness: whether firms react to rivals' price changes within a given response window, and how these reactions vary with firms' demand advantages. I document these response patterns directly in online retail, focusing on algorithmic responsiveness as a channel through which pricing technology can shape market-wide outcomes.

This paper also contributes to the empirical literature on e-commerce price competition. Existing work documents price dispersion, Buy Box competition, and algorithmic repricing within online marketplaces, especially Amazon Marketplace and online retail (Cavallo 2018; Chen, Mislove, and Wilson 2016; He, Hollenbeck, and Proserpio 2022; Lam 2023; Hanspach, Sapi, and Wieting 2024; Musolff 2024). The Project Nessie allegations point to a different object of study: cross-platform pricing responses, in which one online retailer's price changes may affect pricing decisions by other retailers. Studying this setting is difficult because Amazon's prices are relatively visible, while rival retailers' prices are harder to observe at high frequency. I address this gap by constructing synchronized high-frequency pricing data that track identical products across major online retailers. This allows me to measure how price changes propagate sequentially across retailers.

The paper adds to the theoretical literature on algorithmic pricing. Canonical studies show that independently learning algorithms can generate supracompetitive outcomes in simultaneous-move pricing environments (Calvano, Calzolari, Denicolò, and Pastorello 2020; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024). These settings are less suited to markets in which posted prices are observable and firms can react to rivals' posted prices at different times, with demand realized between price adjustments. This paper shifts attention from algorithmic collusion in simultaneous or symmetric environments to systematic algorithmic responsiveness under asynchronous price adjustment.

Related work relaxes symmetry on the supply side, showing that a faster algorithmic firm can use persistent response rules to induce higher prices from a slower or myopic rival (Brown and MacKay

2025). My paper studies a different asymmetry: demand advantage at comparable prices. The key difference is that speed matters here only because of what the responder gains by moving second. A demand-advantaged follower can find it more profitable to match a rival's higher price than to undercut it, while a less demand-advantaged rival would have a stronger incentive to undercut. The mechanism is therefore not commitment to punish a rival, but an incentive-compatible response that the rival can learn.

This paper is also related to the literature on price-matching guarantees (Edlin 1997; Corts 1997; Hviid and Shaffer 1999; Chen, Narasimhan, and Zhang 2001). That literature studies explicit policies under which firms commit to match rivals' lower prices. My setting differs because the response is not an advertised guarantee, but an algorithmic pricing pattern observed in market behavior. The central object is also different: I study why a demand-advantaged firm may find it profitable to match rival price increases, and how rival algorithms can learn from that response.

The paper also contributes to work on sequential price competition and price leadership. Maskin and Tirole (1988) provide a canonical model of alternating-move price competition, in which firms condition their pricing decisions on rivals' current prices. Klein (2021) extends this sequential pricing framework to Q-learning agents. A related empirical literature studies price leadership, showing that dominant firms can use leadership and price experiments to create focal points and coordinate market prices (Byrne and de Roos 2019). In contrast to this emphasis on dominant-firm leadership, the empirical patterns in this paper motivate a focus on dominant-firm followership: Amazon frequently responds to rivals' price changes. To capture this pattern in a learning framework, I extend sequential Q-learning models by introducing empirically grounded demand and timing asymmetries. The resulting framework shows that, when the follower has a demand advantage, moving second can be more valuable than leading.

Finally, the paper connects algorithmic pricing to the literature on demand-side frictions and demand advantages. Search frictions, product differentiation, loyalty, and switching costs can soften price competition by making consumers less willing or able to switch across firms (Hotelling 1929; Diamond 1971; Varian 1980; Narasimhan 1988; Beggs and Klemperer 1992). This literature mainly studies how these frictions affect price competition and market power. I show that they can also shape the roles firms play in sequential algorithmic competition. In my setting, demand advantage matters not only because it weakens consumer substitution, but because it changes which pricing responses become profitable and learnable.

2 Sequential Price Responses Across Platforms

This section builds the empirical case for dominant followership. A platform with a demand advantage, meaning consumers favor it for reasons other than price, such as brand trust, membership programs, or convenience, should be able to price above its rivals without losing much business. Amazon is a natural setting to look for this kind of advantage: it is the dominant platform in this market, allegedly accounting for more than 82 percent of gross merchandise value in 2022 under the online-superstore market definition used in the FTC complaint (Federal Trade Commission 2024). Yet Amazon is not typically the high-price platform; if anything, it is usually priced at or below its rivals. This is the empirical puzzle: why would the platform best positioned to leverage a demand advantage instead compete so aggressively on price? I answer this puzzle by turning from Amazon's price level to its price responses: Amazon's demand advantage shows up not in charging more, but in how it reacts when a rival does. I show that Amazon is unusually likely to raise price soon after a rival raises its own, typically by matching the rival's new price rather than undercutting it. This response pattern lets Amazon benefit from higher prices without ever having to set them first.

2.1 High-Frequency Cross-Platform Pricing Data

Studying price responses requires data that capture a part of competition usually hidden in standard pricing data: the order in which prices change across competing platforms. To recover this order, I build a high-frequency scraping system that tracks identical products across the full set of identified platforms selling each matched variant, collecting prices within synchronized time windows.

Three design features are central. First, I match products at the exact variant level, including SKU, model specifications, and color, so observed price differences reflect platform pricing rather than product heterogeneity. Second, prices for a given product are collected across platforms within the same time window, at roughly two-hour frequency. This synchronization makes it possible to compare prices for the same product across platforms at nearly the same moment. In 91.3% of price-change bins, exactly one eligible platform changes price, so the panel can recover an order of moves rather than only comparing prices at a point in time.[4] Third, for Amazon and Walmart, where the scrape observes marketplace seller identity, I restrict observations to first-party platform listings. This is important because the paper studies platform pricing, not seller pricing within a marketplace. For the other major platforms, the scraped pages do not provide a separate marketplace seller field, so I treat the observed listed price as the platform price.

The panel covers a broad set of consumer electronics and home-appliance categories, including headphones, wearables, smart-home devices, and vacuum and floor-care products, from November 2024 to June 2025.[5] I begin with prominent Amazon-listed products within these categories and keep only products also sold by at least one non-Amazon platform; because the relevant competitors differ across products, the resulting platform set varies and spans both broad-selection online superstores and narrower specialty or brand-specific retailers. After dropping observations with missing prices or marked out of stock, the resulting platform-pricing panel contains 216 matched products, 792 product-platform pairs, and about 1.02 million in-stock price observations. Products are observed on 3.67 platforms on average, and the average scrape interval is 2.3 hours. The product set covers a broad range of price points and demand levels: median product prices range from below \$50 to above \$500, and Amazon's "Bought in Past Month" demand proxy ranges from 50 to 100,000.

2.2 Empirical Puzzle: Amazon's Price Position

If Amazon's demand advantage translated into pricing power in the usual way, it should rarely be the cheapest option on the page. Figure 1 plots Amazon's price relative to the lowest rival price, using observations in which Amazon and at least one rival list the same product.

[4] I define a price update as a change from the platform's most recent observed price for the same product. To avoid treating long gaps as immediate responses, I use only cases where the prior observation occurs within the preceding one to six hours.

[5] Categories are selected based on sufficient price variation to study adjustment behavior.

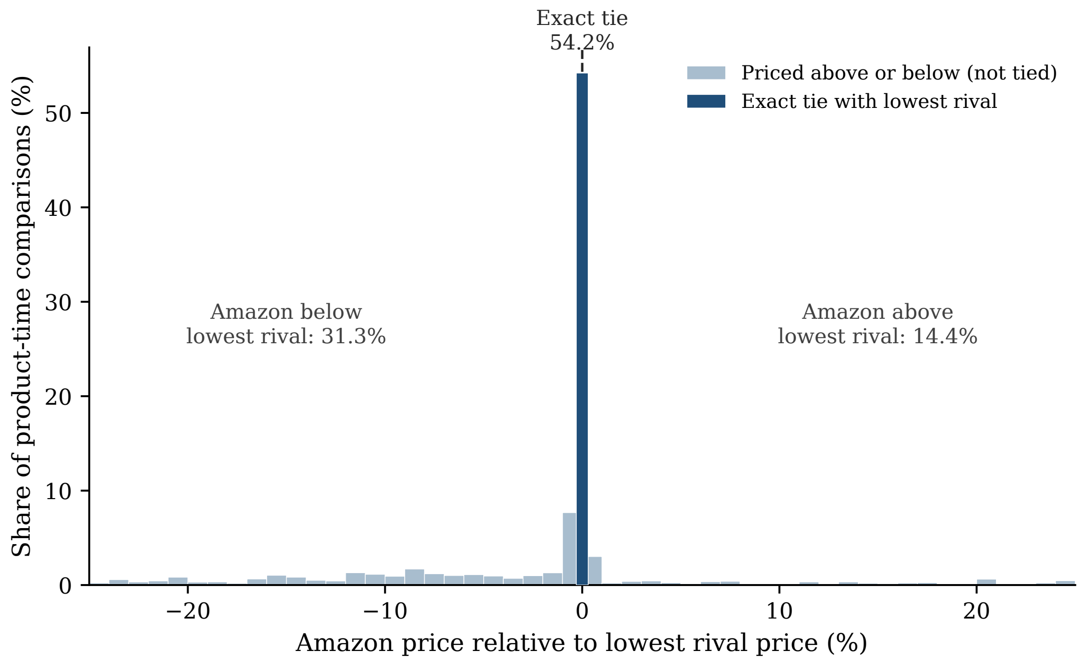

Figure 1. Amazon's Price Relative to the Lowest Rival Price.

Notes: The figure plots Amazon's price relative to the lowest rival price across product-time comparisons in which Amazon and at least one rival have listed prices. A value of zero denotes an exact tie with the lowest rival. Negative values mean Amazon is priced below the lowest rival; positive values mean Amazon is priced above the lowest rival. The figure plots gaps between -25% and 25%; 10.4% of comparisons fall outside this range and are not shown.

Amazon is exactly tied with the lowest rival in 54.2% of comparisons, priced below it in 31.3%, and priced above it in only 14.4%; more than three quarters of comparisons fall within 10% of the lowest rival price. This pattern is stable over time: Amazon's weekly lowest-price share ranges from 77.5% to 88.9% across the sample period, consistently above the weekly mean lowest-price share among major rivals (Walmart, Target, Best Buy, and Home Depot), which ranges from 40.4% to 60.2%.

Amazon's distinctive market role, then, is not expressed through a persistently higher price. It shows up instead in how Amazon responds when rivals move first.

2.3 Amazon Is More Likely to Raise Price After Rival Increases

Amazon is unusually likely to raise price soon after a rival does. Table 1 compares, for each Amazon-product observation, moments when a rival has just raised price against moments when no rival has, using whether Amazon raises price within the next 48 hours as the outcome.[6] Amazon increases within 48 hours in 10.2% of cases when no rival has just raised price, compared with 30.7% of cases when a rival has. The difference is 20.4 percentage points. The same pattern remains after controlling for product, platform, week, and day-of-week differences: Amazon is 15.9 percentage points more likely to raise price after a rival increase ($p < .001$; Panel A of Appendix Table A.1). The broad response pattern therefore shows that rival increases predict later Amazon increases.

Table 1. Raw Increase Probabilities After Rival Price Increases

Platform	P(Increase within 48h | No rival increase)	P(Increase within 48h | Rival increase)	Difference (p.p.)
Amazon	10.2	30.7	20.4
Walmart	8.5	13.5	4.9
Target	3.5	9.7	6.2
Best Buy	7.5	9.6	2.1
Home Depot	5.9	15.2	9.3

Notes: Entries report raw percentages in the main response-regression sample. The first column reports the probability that the focal platform increases its price within 48 hours when no rival increases price at time t. The second column reports the same probability after at least one rival selling the same product increases price at time t.

This broad pattern does not by itself show where the pricing sequence begins. A rival increase may already be a response to an earlier price increase by another platform. I therefore restrict attention to isolated rival increases, in which the ordering is unambiguous: no platform raised price for the product in the prior 48 hours, exactly one identified rival platform (for Amazon, any identified non-Amazon platform selling the product) then raises price, and the responding platform's own price is still unchanged at that moment.[7]

Table 2 establishes this follower pattern in three steps. First, Amazon follows rivals. After an isolated rival increase, Amazon is 12.9 percentage points more likely to raise its own price. Second, Amazon follows more strongly than other platforms generally do. The corresponding response among the four major non-Amazon platforms is only 3.7 percentage points, yielding a gap of 9.3 percentage points ($p < .001$). Expanding the comparison to all non-Amazon platforms leaves the conclusion unchanged: their pooled response is 4.2 percentage points, and the gap relative to

[6] Among the response horizons I consider, the 48-hour window captures the largest share of later same-product adjustments, 33.9%.

[7] The analogous definition applies to isolated rival decreases.

Table 2. Responses to Isolated Rival Price Increases

	Price increase
Response to isolated rival price increase	
Amazon	12.94
	(1.79)
Non-Amazon major platforms	3.65
	(0.95)
Difference: Amazon − non-Amazon	9.29
	(1.73), $p < .001$
Observations	597,334
All-platform robustness	
All non-Amazon platforms	4.24
	(0.95)
Difference vs. Amazon	8.54
	(1.77), $p < .001$
Total observations	856,436
Response to isolated Amazon price increase	
Non-Amazon major platforms	0.15
	(0.73), $p = .837$
	$N = 406{,}877$
All non-Amazon platforms	−0.36
	(0.65), $p = .581$
	$N = 665{,}979$

Notes: Entries are percentage-point effects on increasing price within 48 hours after an isolated price increase, relative to otherwise comparable periods without one. The Amazon row uses isolated non-Amazon increases facing Amazon; the non-Amazon row pools Walmart, Target, Best Buy, and Home Depot when they face isolated increases by their own rivals. The difference row tests equality between these two coefficients, with its standard error, clustered by product, in parentheses. The all-platform robustness block re-estimates this comparison using every non-Amazon platform in the pricing panel, rather than only the four majors; the corresponding Amazon coefficient in that expanded sample is 12.78 percentage points with standard error 1.82. The bottom panel reverses direction, reporting non-Amazon platforms' response to an isolated Amazon price increase, first pooling only the four majors and then every non-Amazon platform in the panel; both coefficients are small and statistically indistinguishable from zero. Standard errors are reported in parentheses; for difference rows, the parenthesized value is the standard error of the difference. All specifications include product, platform, week, and day-of-week fixed effects; standard errors are clustered by product.

Amazon remains 8.5 points.

Third, rivals do not similarly follow Amazon. When Amazon is the isolated price increaser, the four major non-Amazon platforms show essentially no response: their response estimate is 0.15 percentage points ($p = .837$). Pooling all non-Amazon platforms gives a similarly small and statistically indistinguishable estimate of -0.36 percentage points ($p = .581$). Thus rival increases predict Amazon increases, but Amazon increases do not predict rival increases.

This directional asymmetry helps address a common-cost-shock explanation. If isolated price increases mainly reflected product-level shocks affecting all platforms selling the same product, response probabilities should not differ so sharply by the identity of the responding platform. Panel B of Appendix Table A.1 shows that the pattern holds firm by firm: none of Walmart, Target, Best Buy, or Home Depot approaches Amazon's response individually.

The isolated-event response is also strongest among higher-demand products: it rises from 8.3 percentage points among low-demand products to 10.9 among middle-demand products to 15.5 among high-demand products (Appendix Table A.2). This pattern motivates the model below. When more demand is at stake, responding to a rival's price increase can be more valuable for a demand-advantaged platform. A continuous version of this test, interacting the treatment with log product demand, points in the same direction (1.9 percentage points per log-point of demand, $p = .057$).

Amazon's outsized response to rivals is also robust to the choice of response window. Restricting to isolated rival price increases, Amazon's response remains significantly larger than the pooled response of the other major platforms across alternative response windows: the gap is 4.4 percentage points at 12 hours ($p = .001$), 5.3 at 24 hours ($p = .001$), 9.3 at 48 hours ($p < .001$), and 12.0 at 72 hours ($p < .001$). The pattern also holds across product categories, as shown in Appendix Figure A.1.

2.4 Amazon Raises Price Soon After Rival Increases

Amazon is unusually likely to raise price after isolated rival increases. The next question is whether this pattern is tied to the rival's move. If Amazon were already on an upward path, the same events should predict Amazon increases before the rival move. If the estimate only reflected Amazon's frequent repricing, Amazon's increases should be spread throughout the response window rather than concentrated soon after the rival move.

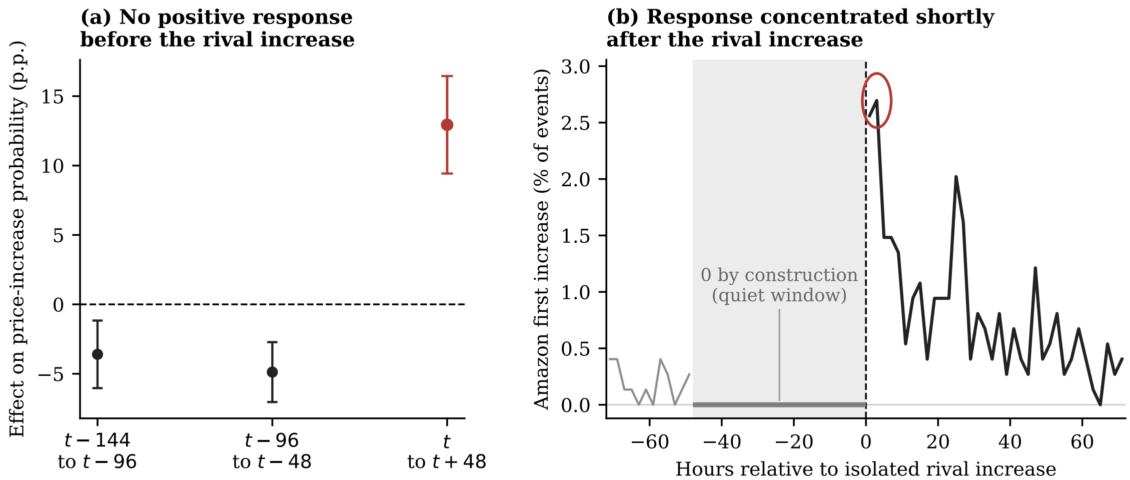

Figure 2. Timing Evidence Around Isolated Rival Price Increases.

Notes: Panel (a) reports Amazon's percentage-point effect on increasing price within 48 hours of an isolated rival increase. Placebo windows assign the same rival-increase events to fake event times 96 or 144 hours earlier. The placebo specifications include product, week, day-of-week, and platform fixed effects, with standard errors clustered by product. Panel (b) conditions on isolated rival increases facing Amazon and plots the share of events for which Amazon's first subsequent price increase occurs in each two-hour bin; the vertical dashed line marks the isolated rival increase. The quiet window is zero by construction because the isolated event requires no same-direction price increase for the product during the prior 48 hours.

Figure 2 addresses both alternatives. Panel (a) shows no comparable positive response in the fake earlier windows. Panel (b) shows that Amazon's first increase is concentrated shortly after the rival move. Of Amazon's responses within the window, 28 percent occur within the first 8 hours, with no anticipatory movement beforehand. The timing evidence therefore supports the follower interpretation: Amazon is not simply continuing a prior price increase or repricing often in general; it is raising price after the rival's observed increase.

2.5 Amazon Matches When Responding to Rival Price Increases

Amazon raises price soon after isolated rival increases. The next question is where Amazon's response lands. If Amazon matches the rival's new price, the higher price stands. If Amazon remains below the rival, Amazon's lower price limits the increase. I therefore measure Amazon's response price relative to the rival's new price.

I classify Amazon's first price increase within 48 hours as below, at parity with, or above the rival's new price. For example, if Walmart raises the price of a headphone from \$98 to \$109 and

Amazon later moves from \$98 to \$109, I classify Amazon's adjustment as a parity response. If Amazon moves to \$108, I classify it as below the rival's new price; if it moves to \$110, I classify it as above.

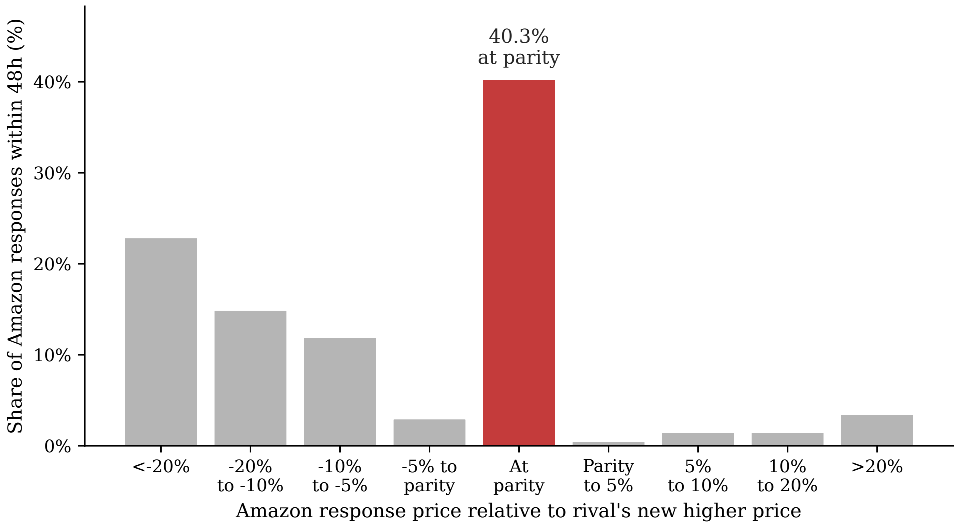

Figure 3. Amazon's Response Price Relative to the Rival's New Price.

Notes: The figure conditions on isolated rival price increases facing Amazon for which Amazon raises price within 48 hours. It plots Amazon's response price relative to the rival's new price. Gaps are measured in percentage points relative to the rival's new price; the parity bar includes responses classified as matches under the tolerance described in the text.

The largest mass in Figure 3 is at parity: 40.3% of Amazon's responses match the rival's new price. Including responses below the rival's new price, 93.0% are at or below the rival's new price. The important point is not price matching by itself, but the matching response of Amazon, the demand-advantaged platform, after a rival moves first. A substantial share of these responses preserve the rival's higher price rather than undoing it through undercutting.

2.6 Rival Increases Last Longer When Amazon Matches

I next ask whether rival increases last longer when Amazon matches them, and whether the same pattern holds for rival decreases. For each isolated rival price change facing Amazon, I compare two cases: Amazon matches the rival's new price within 48 hours, or Amazon does not match it within 48 hours. I then track whether the rival's price remains on the new side of the price change over the following 60 days. If Amazon's matching response helps preserve the rival's higher price, then rival increases followed by an Amazon match should last longer than rival increases not followed by an Amazon match.

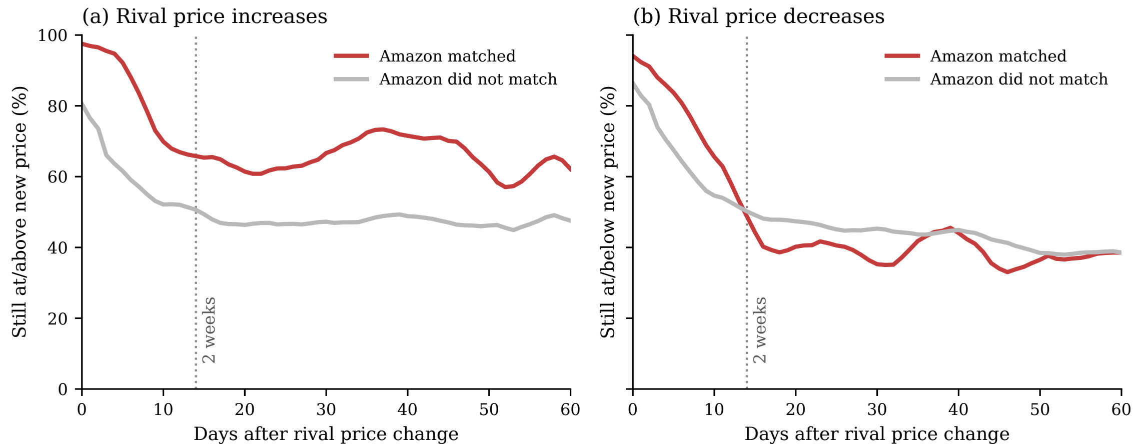

Figure 4. Persistence of Rival Price Changes by Amazon's Response.

Notes: The figure conditions on isolated rival price changes facing Amazon. Panel (a) compares rival increases that Amazon matches within 48 hours with rival increases that Amazon does not match within 48 hours. Panel (b) makes the analogous comparison for rival decreases. Lines report 5-day centered moving averages of daily persistence shares. For rival increases, persistence means that the rival's later price remains at least as high as the new price set immediately after the increase. For rival decreases, persistence means that the rival's later price remains no higher than the new price set immediately after the decrease. For example, a $100-to-$110 increase is counted as persistent only if the later rival price is at least $110; a $100-to-$90 decrease is counted as persistent only if the later rival price is no higher than $90.

The raw checkpoint values show the same pattern. At two weeks, 67.1% of matched rival increases remain at or above the new price, compared with 51.3% of unmatched increases; the gap remains at 60 days (63.0% versus 47.1%). Product-clustered checkpoint tests show that the increase gap is statistically significant at two weeks (15.8 percentage points, $SE = 5.8$, $p = .007$) and remains positive at 60 days (15.9 percentage points, $SE = 8.3$, $p = .058$). Rival decreases show the opposite pattern: matched decreases persist less often than unmatched decreases at two weeks (44.6% versus 49.2%), though this gap narrows by 60 days (37.7% versus 38.6%). These decrease gaps are not statistically significant (-4.5 percentage points, $SE = 5.0$, $p = .361$ at two weeks; -0.9 percentage points, $SE = 7.2$, $p = .895$ at 60 days). Amazon's matching response is therefore associated with more durable higher rival prices specifically, not with durable rival price changes in general.

The next section builds a model to explain why following can be incentive-compatible for a demand-advantaged firm.

3 Dominant Followership

When two undifferentiated firms compete only on price, the standard benchmark is the simultaneous-move Bertrand model, which predicts prices equal to marginal cost. Sequential timing alone does not overturn this intuition: if firms are otherwise symmetric and can undercut each other, the force pushing prices toward marginal cost remains. The standard intuition changes once the model is modified in a seemingly minor way: one firm has a demand advantage at price parity. Instead of assuming that firms are identical in every respect, I allow one firm, hereafter the dominant firm, to capture more than half of market demand when prices are equal. This assumption can be microfounded by a simple lexicographic decision rule. Consumers first compare prices, but when prices are equal or perceived as equivalent, they choose based on firm identity, brand, convenience, or other non-price advantages that favor the dominant firm. This advantage need not operate through the dominant firm charging high prices itself or moving first to claim the usual Stackelberg advantage: the relevant margin is the follower's response incentive when a rival moves first.

The analysis builds the mechanism in steps. I first isolate the core mechanism in a one-clearing pricing game: demand advantage raises prices only when the demand-advantaged firm is the follower, since only then does the advantage turn the follower's incentive from undercutting to matching. Appendix B.6 shows that this mechanism survives when firms retain loyal consumers at higher prices. I then add an interim market clearing, introducing a dynamic tradeoff between the short-run cost of initiating a price increase and the future value of the dominant firm's response. This tradeoff depends on how reliably the dominant firm can be counted on to adjust its price after the rival adjusts its own. The infinite-horizon extension asks whether the resulting high price persists once the game continues, rather than only for the one extra period a finite game can capture. Once the high matched price is reached, the rival can always undercut it; what deters this ongoing temptation is the rival's own response reliability.

3.1 Environment

Suppose there are two firms, indexed by $i \in \{D, S\}$. Firm D is the dominant, demand-advantaged firm, and firm S is the less demand-advantaged rival. Firms choose prices from a finite evenly spaced grid

$$\mathcal{P} = \{\delta, 2\delta, \dots, M\delta\},$$

where $\delta > 0$ is the grid step and $M\delta$ is the maximum price. Let $p_i \in \mathcal{P}$ denote firm i's price. For any x, define the grid floor

$$\lfloor x \rfloor_{\mathcal{P}} = \max\{p \in \mathcal{P} : p \le x\}.$$

For numerical illustrations, I normalize the maximum price to one and use a five-point grid, so $\delta = 0.2$ and $\mathcal{P} = \{0.2, 0.4, 0.6, 0.8, 1\}$.

The grid reflects the discrete price changes observed in the data.[8]

Total demand is normalized to one. Demand is allocated according to

$$q_D(p_D, p_S) = \begin{cases} 1, & p_D < p_S, \\ \lambda, & p_D = p_S, \\ 0, & p_D > p_S, \end{cases} \qquad q_S(p_D, p_S) = 1 - q_D(p_D, p_S).$$

The low-price firm receives the entire market. When prices are equal, the dominant firm receives share $\lambda \in [1/2, 1]$, with $\lambda = 1/2$ corresponding to the symmetric benchmark. I treat λ as capturing non-price platform advantages, such as brand trust, membership programs, or switching frictions, rather than an advantage created within the model by current-period pricing.

Timing and market clearing. The baseline analysis compares two standard one-shot timing benchmarks: simultaneous-move Bertrand competition, in which firms set prices without observing each other's current choice, and sequential-move Bertrand competition (Stackelberg timing), in which one firm, the leader, sets its price first and the other, the follower, observes it before responding. In both cases, the market clears only once, after both firms have set their prices, so there is no demand or payoff before the full price pair is determined. This one-shot structure means the game ends with zero probability of either firm getting a further opportunity to respond. With

[8] It can also be interpreted as a reduced-form representation of limited consumer price perception: under Weber's law, the smallest noticeable change in a stimulus is proportional to its initial level, so sufficiently small relative price differences may generate the same perceived price and demand response (Fechner 1860; Monroe 1971, 1973). Observed pricing patterns are consistent with discrete, economically meaningful price steps: average dollar adjustment sizes are \$5.17 for products under \$50, \$13.08 for products between \$50 and \$150, \$28.55 for products between \$150 and \$500, and \$81.64 for products above \$500.

marginal cost normalized to zero, realized profit is

$$\pi_i(p_D, p_S) = p_i q_i(p_D, p_S).$$

The two-clearing and infinite-horizon extensions below relax this zero-response-probability assumption, introducing timing friction as a force distinct from demand asymmetry.

3.2 One-Clearing Pricing Games

Matching versus undercutting. The key mechanism can be seen from a firm's best response to a rival price $p \in \mathcal{P}$. On the price grid, the relevant local choice is whether to match p or undercut it by one step, $p - \delta$, since any lower price also receives the entire market but yields a lower margin. If the firm receives demand share s at price parity, matching yields sp, while undercutting yields $p - \delta$. The firm is therefore willing to match if and only if

$$sp \geq p - \delta,$$

or equivalently,

$$p \leq \frac{\delta}{1 - s}. \tag{1}$$

The matching cutoff $\delta/(1 - s)$ is increasing in the firm's demand share at parity.

Simultaneous pricing. Consider a candidate common-price equilibrium $p_D = p_S = p$ under simultaneous pricing. Applying condition (1), the dominant firm's no-undercutting condition is

$$p \leq \frac{\delta}{1 - \lambda},$$

while the rival's condition is

$$p \leq \frac{\delta}{\lambda}.$$

Under demand asymmetry, $\lambda > 1/2$, the rival's condition is more restrictive. Hence the highest common-price equilibrium is

$$p^{Sim}(\lambda) = \left\lfloor \frac{\delta}{\lambda} \right\rfloor_{\mathcal{P}}.$$

Thus, demand asymmetry by itself is not enough to raise prices in the simultaneous benchmark. The common price is disciplined by the rival, which has the stronger incentive to undercut.

Sequential pricing. Sequential pricing changes the constraint because the follower observes the first mover's price before choosing. This one-clearing timing structure gives the first mover no opportunity to revise again before demand is realized. The highest sustainable matched price is therefore pinned down by the follower's willingness to match.

Let p_D^{follow} denote the highest matched price sustainable when the dominant firm follows, and let p_S^{follow} denote the corresponding price when the rival follows.

Proposition 1 (Dominant followership raises equilibrium prices)**.** *Suppose* $\lambda > 1/2$. *The highest matched price sustainable under sequential pricing satisfies*

$$p_D^{follow} \geq p_S^{follow}.$$

[9] *Specifically,*

$$p_D^{follow} = \left\lfloor \frac{\delta}{1-\lambda} \right\rfloor_{\mathcal{P}},$$

while

$$p_S^{follow} = \left\lfloor \frac{\delta}{\lambda} \right\rfloor_{\mathcal{P}}.$$

The result follows from applying the no-undercutting condition (1) to the follower: $\lambda > 1/2$ implies $\delta/(1-\lambda) > \delta/\lambda$. Figure A.2 plots the comparative statics.

Table 3 illustrates the result in a five-point price grid.

[9] The inequality is weak because prices are chosen from a discrete grid; it is strict whenever $\mathcal{P}$ contains a price in $(\delta/\lambda, \delta/(1-\lambda)]$.

Table 3. Matched Equilibrium Prices: An Illustration

Demand environment	Simultaneous pricing	Sequential $D \to S$	Sequential $S \to D$
Symmetric demand, $\lambda = 1/2$	2δ	2δ	2δ
Asymmetric demand, $\lambda = 0.7$	δ	δ	3δ

Notes: Entries report the highest matched equilibrium price on the normalized five-point grid. In sequential pricing, $D \to S$ denotes the dominant firm moving first and the rival following; $S \to D$ denotes the reverse.

The table shows that equilibrium prices rise only when the two forces coincide. Demand asymmetry alone is not enough: under simultaneous pricing or when the dominant firm leads ($D \to S$), the rival's no-undercutting condition binds either way, so the price is no higher than under symmetric demand. Sequential timing alone is not enough either: without demand asymmetry, all three timing structures give the same matched price. The price rises only when asymmetric demand is paired with the dominant firm following ($S \to D$). In that case, the same price increase creates opposite response incentives: the dominant firm is willing to match because it keeps substantial demand at comparable prices, while the rival has a stronger incentive to undercut. Appendix B.2 formalizes this behavioral separation and shows that both firms can prefer the dominant-followership role assignment among matched sequential outcomes.

3.3 Costly Rival Initiation

The one-clearing game abstracts from the current cost of initiating a price increase: if the dominant firm matches, demand is realized only after both firms are at the higher price. I next ask whether dominant followership can still support rival initiation when initiation is costly. I introduce this cost by giving the dominant firm one extra chance to revise its price, but only after the market has already cleared once. Because the market clears before that final revision, the rival may lose current demand before the dominant firm matches. The key question is whether the future value of the dominant firm's match is large enough to offset the rival's current demand loss.

I write this timing as $i \to j \to M \to i \to M$, where M denotes a market clearing. Let a_1 denote firm i's initial price, a_2 firm j's response, and a_3 firm i's final price. The two market clearings occur at price pairs (a_1, a_2) and (a_3, a_2), so firm k's total payoff is

$$U_k(a_1, a_2, a_3) = \pi_k(a_1, a_2) + \beta\pi_k(a_3, a_2), \qquad k \in \{i, j\}, \tag{2}$$

where $\beta \in (0, 1]$ weights the second market. I solve this finite game by backward induction.

Proposition 2 (A threshold for costly rival initiation)**.** *Consider the order $D \to S \to M \to D \to M$. Fix the dominant firm's initial price at δ*[10]*, and let $p^M \equiv \lfloor \delta/(1-\lambda) \rfloor_{\mathcal{P}}$ denote the highest price the dominant firm will match at the final revision. Whenever $p^M > \delta$, the rival's best response is to raise price to p^M if and only if*

$$\beta > \bar{\beta}(\lambda) \equiv \frac{\delta}{p^M - \delta}.$$

The threshold $\bar{\beta}(\lambda)$ is weakly decreasing in λ: a larger demand share at parity requires less patience to make rival initiation worthwhile.

Corollary 4 in Appendix B.3 verifies that this threshold is strong enough to support a complete equilibrium, not only a conditional response rule.

The result shows when costly initiation can occur. The rival is willing to raise price first only when it is sufficiently forward-looking: it must value the future matched price enough to compensate for the current demand it gives up. The firm that initiates the increase need not be the firm that makes it sustainable. The identity of the final responder is therefore essential: if the rival receives the final revision opportunity instead, its stronger incentive to undercut prevents the high price from being sustained. Appendix B.4 provides this reverse-timing comparison.

Figure 5 plots the threshold directly. As λ rises, the dominant firm becomes willing to match a higher price, $p^M(\lambda)$. A higher p^M raises the rival's payoff after the dominant firm matches, so initiation is profitable at a lower discount factor.

[10]The grid's lowest price is a natural starting point for this illustration because it is the empirically relevant case: Amazon is typically at or near the lowest price among competitors (Figure 1: tied with the lowest rival in 54.2% of comparisons, priced above it in only 14.4%), though the grid floor here is a modeling device and not a claim that some absolute lowest price exists in practice.

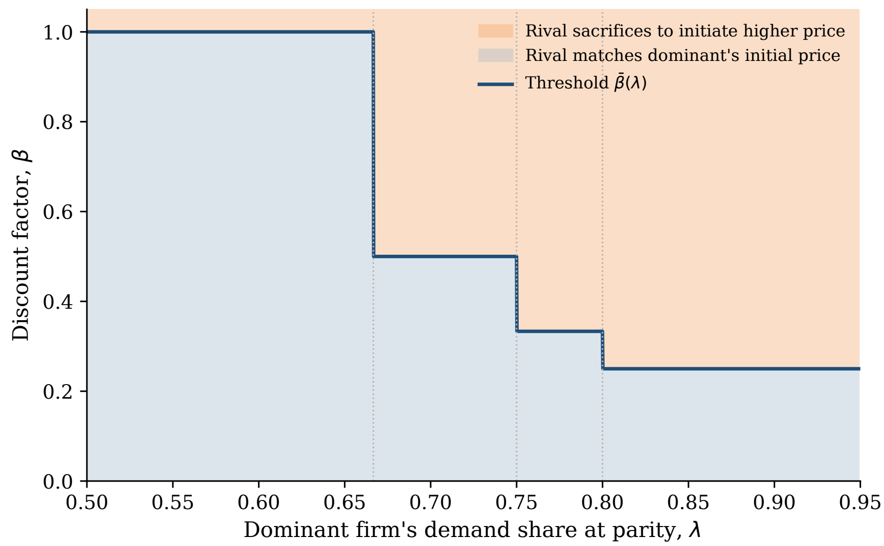

Figure 5. Threshold for Rival Price Initiation.

Notes: The figure uses the normalized five-point grid for the $D \to S \to M \to D \to M$ timing in Proposition 2, fixing the dominant firm's initial price at δ. The line plots $\bar{\beta}(\lambda) = \delta/(p^M(\lambda) - \delta)$, the exact threshold above which the rival's unique best response is to raise price to $p^M(\lambda)$ rather than match δ. Dotted vertical lines mark the grid points at which $p^M(\lambda)$ steps up.

Proposition 2 assumes that the dominant firm's final response is guaranteed. I next relax this by introducing response reliability. Let $\rho_i \in [0, 1]$ denote the probability that firm i can revise its price when its turn immediately follows a rival price change (illustrated in Figure 8). If the rival has not changed price, the firm can revise normally, without ρ_i entering. In the two-clearing game, the dominant firm's final turn is therefore gated only if the rival moves away from the shared starting price, $a_2 \neq a_1 = \delta$.[11] With probability ρ_D, the dominant firm revises optimally; with probability $1 - \rho_D$, it does not revise and its price remains at δ.

Corollary 1 (Response reliability and costly initiation)**.** *Under the timing of Proposition 2, suppose the dominant firm's final turn is gated only after a rival price change. Then the rival's threshold for costly initiation is*

$$\bar{\beta}(\lambda, \rho_D) \equiv \frac{\delta}{\rho_D \, p^M(\lambda) - \delta},$$

which equals $\bar{\beta}(\lambda)$ when $\rho_D = 1$. If $\rho_D \, p^M(\lambda) \leq \delta$, no $\beta < 1$ satisfies this threshold, and costly initiation cannot be rationalized at any discount factor.

[11] In this finite game, $a_2 \neq a_1$ is the analogue of a rival price change. The infinite-horizon game and the learning simulations use the same response-window rule: the window is triggered when the rival changes its posted price relative to its previous posted price.

Patience and dominant-firm response reliability are therefore complements at the initiation stage. The threshold $\bar{\beta}(\lambda, \rho_D)$ is the minimum patience required for the rival to raise price first, so a lower threshold means initiation is easier. A higher ρ_D lowers this threshold by making the future match more likely; a sufficiently unreliable dominant firm forecloses costly initiation regardless of how patient the rival is.

Figure 6 plots $\bar{\beta}(\lambda, \rho_D)$ directly against ρ_D, holding λ fixed at a few representative values. The threshold falls monotonically as ρ_D rises, and diverges as ρ_D approaches the point at which $\rho_D\, p^M(\lambda) = \delta$ from above; at $\rho_D = 1$, each line reproduces the corresponding step of Figure 5.

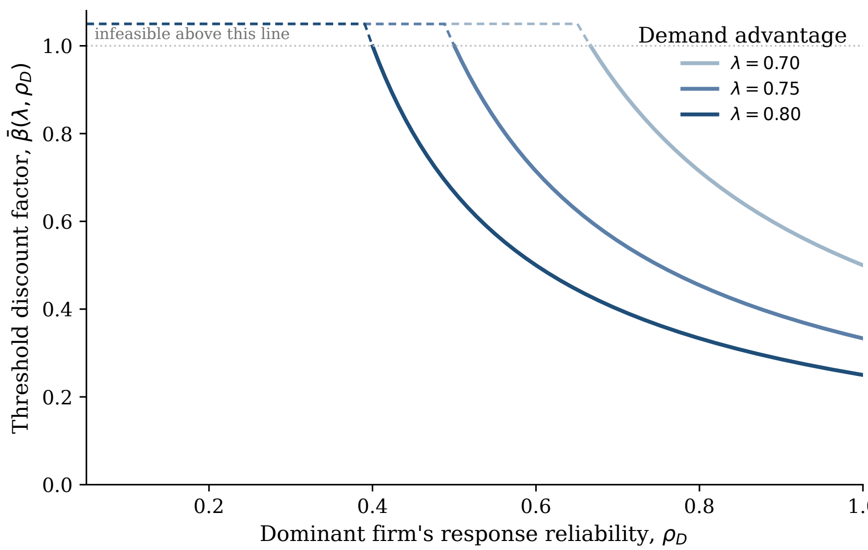

Figure 6. Threshold for Rival Price Initiation, by Response Reliability.

Notes: Each line plots $\bar{\beta}(\lambda, \rho_D) = \delta/(\rho_D\, p^M(\lambda) - \delta)$ for a fixed λ, capped at 1 for display where the threshold is infeasible (dashed). At $\rho_D = 1$, each line's value equals the corresponding step of Figure 5.

3.4 Infinite-Horizon Markov Extension

The finite-game extension asks whether a rival can be willing to initiate a higher price when doing so is costly in the current period. I next ask whether the resulting high matched price can persist once the game continues. The new issue is defection: after the market reaches the high matched price, the rival may undercut whenever it gets another chance to move.

Consider the two-price case $\mathcal{P} = \{L, H\}$, where $0 < L < H$. Firms alternate revision opportunities on a fixed clock, D then S then D then S. I use the same response-window rule as above. A firm moves normally unless its turn immediately follows a rival price change. In that case, it can revise only with probability ρ_i; otherwise prices remain unchanged for that period. The Markov state is (p_D, p_S, m, c): the current posted-price pair, the firm $m \in \{D, S\}$ whose turn is next, and an indicator c for whether the preceding period involved a price change. For the analytical result, I fix $\rho_D = 1$, so the dominant firm is fully reliable, and vary ρ_S. This isolates the rival's own response reliability: how quickly it can re-establish a high price after giving one up.

Section 3.3 showed why high ρ_D matters for initiation: the rival is willing to raise price only if the dominant firm's later match is likely enough. The role of ρ_S below is different. It governs the rival's own later response opportunities after the high price has already been matched, and therefore affects the temptation to undercut rather than the incentive to initiate. I therefore set $\rho_D = 1$ and verify a candidate policy in which the rival initiates the high price from (L, L), the dominant firm matches, and both firms remain at (H, H):

price state (p_D, p_S)	$\sigma_D(p_D, p_S)$	$\sigma_S(p_D, p_S)$
(H, H)	H	H
(H, L)	L	H
(L, H)	H	H
(L, L)	L	H

where $\sigma_i(p_D, p_S)$ is firm i's price choice when it receives the revision opportunity.

The relevant price region is the same matching-versus-undercutting wedge as Section 3.2: the dominant firm is willing to match H, while the rival would undercut H if it could immediately exploit the matched price. Appendix B.5 verifies that the rival prefers preserving H to undercutting it, along with the initiation and off-path checks, by solving the Bellman equations.

Proposition 3 (Dominant followership can persist in repeated competition)**.** *Suppose $\lambda > 1/2$, $\rho_D = 1$, and*

$$(1 - \lambda)H < L < \lambda H.$$

There is a nonempty parameter region in which a Markov-perfect equilibrium has the rival initiate the high price, the dominant firm match, and both firms remain at (H, H) indefinitely. This region weakly expands as the rival's response reliability ρ_S falls.

The new force in the repeated game is not initiation but persistence: once the market reaches (H, H), the rival may be tempted to undercut. Lower ρ_S makes that temptation easier to deter. If the rival undercuts, it must later use its own response opportunity to climb back to H; when ρ_S is low, that recovery is delayed, so the rival gives up more future high-price profit. Figure 7 illustrates this logic. After defecting, a moderately reliable rival ($\rho_S = 0.5$) returns to H faster than a fully unreliable rival ($\rho_S = 0$), holding $\rho_D = 1$ fixed throughout as in the proposition. The shaded gap is the rival's own cost of undercutting.

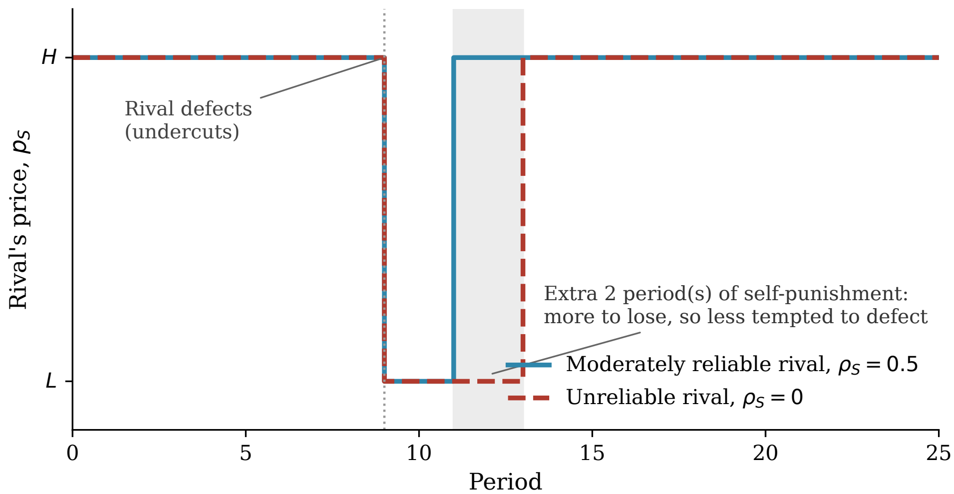

Figure 7. A Rival's Defection Costs Itself More When It Is Unreliable

Notes: Simulated path of the rival's own price under the candidate policy and the response-window rule in Section 3.4, with $\rho_D = 1$ throughout and the same sequence of underlying random draws for both lines, so only ρ_S differs between them. The rival is forced to defect at period 9; play afterward follows the candidate policy and the stochastic response-window rule with no further intervention. This illustrates the mechanism behind Proposition 3 rather than proving it; the incentive conditions themselves are verified directly in Appendix B.5.

The matching pattern from Section 3.3 therefore does not unravel once the game keeps going. Lower ρ_S makes undercutting less attractive for the rival, making the high price easier to sustain. Together, Corollary 1 and Proposition 3 motivate the asymmetric-response treatment in Section 4: $\rho_D = 1$ makes costly initiation more attractive, while $\rho_S = 0$ makes the matched price more persistent.

4 Algorithmic Learning with Dominant Followership

The analysis so far shows why dominant followership can be privately optimal when firms understand the incentives they face. In online retail, that is a strong assumption. Firms set prices across large product assortments, while demand conditions and rival response patterns vary across products and change in real time. A pricing algorithm may therefore have to learn the mechanism from market feedback rather than solve it from a known payoff model. I ask whether Q-learning algorithms can do so. Existing Q-learning models of algorithmic pricing typically study symmetric firms: either firms move simultaneously (Calvano, Calzolari, Denicolò, and Pastorello 2020; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024) or they alternate sequentially with symmetric response opportunities (Klein 2021). The empirical evidence and the model point to a different learning environment. First, firms set prices sequentially: one platform's price change can become visible before a rival adjusts, and demand can be realized against the posted price in the interim. Second, response reliability is asymmetric: Amazon responds to rival price changes far more reliably than other major platforms do. Third, Proposition 1 shows that demand asymmetry itself changes a firm's incentive to match rather than undercut when it responds, so whether prices rise depends on the interaction between who holds the demand advantage and who responds reliably, not on either alone. A learning environment that treated pricing as simultaneous, gave every firm the same response reliability, or fixed demand shares symmetrically would erase exactly the mechanism these facts and this result identify. I therefore let demand advantage and response reliability both vary and interact.

The results then proceed in three steps. I first ask when learned prices rise and show that demand asymmetry raises prices only when it is paired with asymmetric response timing. I then use one-learner decompositions to identify whose learning sustains the price increase. Finally, I show that the learning that raises prices and the gains from those prices need not accrue to the same firm. This distinction bears directly on the Project Nessie allegation: it need not be the dominant firm's own algorithm that is sophisticated for it to benefit from a rival's learned price increases.

4.1 Environment and Learning Rule

Environment. I embed the demand environment from Section 3 in a repeated learning problem with two firms, D and S, both choosing prices from the same grid $\mathcal{P}$. Each firm's problem is a Markov decision process:

State. $s_{i,t} = (p_{i,t-1}, p_{j,t-1})$, $j \neq i$: the inherited posted-price pair firm i observes when it has a pricing opportunity. Appendix Section A.2 supports this two-price state empirically.

Action. $a_{i,t} \in \mathcal{P}$: the price firm i chooses from the common grid.

Reward. $\pi_i(p_{i,t}, p_{j,t})$: firm i's realized profit under the demand environment in Section 3.1, once both firms' posted prices are set for the period.

Timing. In the posted-price treatment, at most one firm adjusts its price in a period, and demand is realized against the resulting posted prices. The next turn then depends on whether the active firm changed its price. If firm i does not change price, firm j's scheduled turn proceeds normally. If firm i changes price, firm j's scheduled turn becomes a response window: with probability ρ_j, firm j can revise; with probability $1 - \rho_j$, no firm moves and the market clears again at unchanged prices. Thus, ρ_j governs response reliability after rival price changes, not ordinary pricing turns. This differs from Klein (2021), where firms alternate deterministically with symmetric response opportunities. Here, ρ_D and ρ_S can differ, letting the simulations reflect the asymmetric response reliability documented above. Figure 8 summarizes the rule.

A separate simultaneous-updating benchmark shuts down this response channel entirely: both firms choose prices before demand is realized, so neither observes or responds to the rival's current move. I use it below to isolate the demand effect alone.

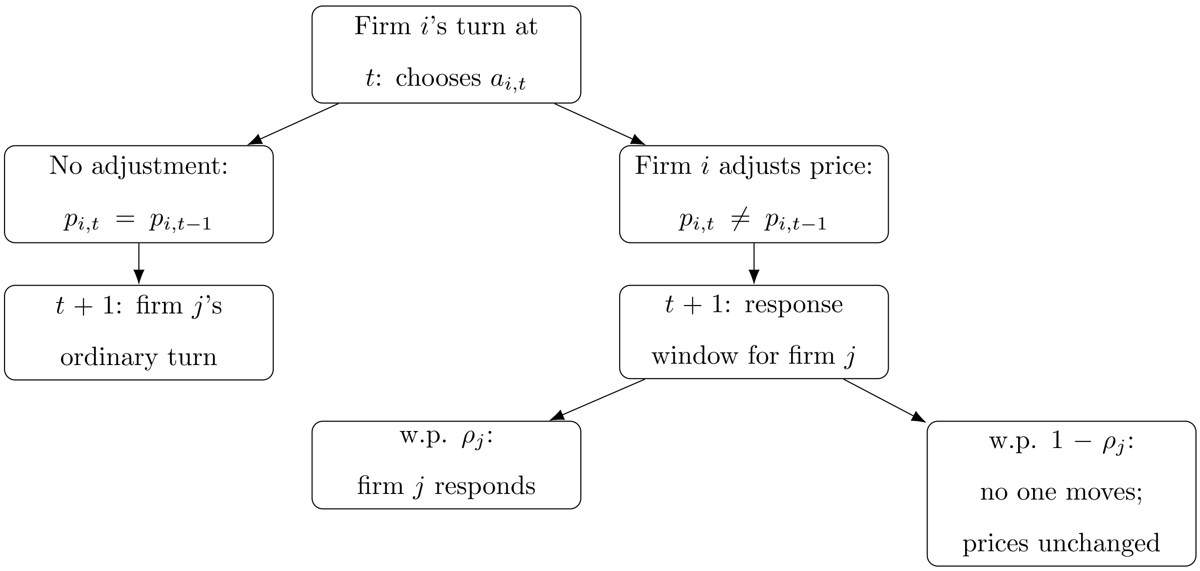

Figure 8. Response-Window Timing.

Notes: The figure illustrates the response-opportunity rule. After firm i's pricing opportunity at t, whether firm j's next scheduled turn is ordinary or gated depends on whether firm i changed its price. In a response window, firm j revises only with probability ρ_j.

Learning rule. Firms use standard Q-learning with ϵ-greedy exploration. Each firm maintains a table $Q_{i,t}(s, a)$, its estimate of the discounted value of price a in state s, and updates it each period as summarized in Algorithm 1.

Algorithm 1 Q-Learning Update for Firm i in Period t

Observe state $s_{i,t} = (p_{i,t-1}, p_{j,t-1})$
Draw $u_t \sim U[0,1]$
if $u_t \leq 1 - \epsilon_t$ **then** ▷ exploit
    $a_{i,t} \leftarrow \arg\max_{a \in \mathcal{P}} Q_{i,t}(s_{i,t}, a)$
else ▷ explore
    $a_{i,t} \leftarrow$ price drawn uniformly at random from $\mathcal{P}$
end if
Post price $p_{i,t} \leftarrow a_{i,t}$
if simultaneous timing **then**
    $\widehat{Q}_{i,t} \leftarrow \pi_i(p_{i,t}, p_{j,t}) + \beta \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+1}, a')$
else ▷ response-window timing
    Accumulate discounted profit over periods $t, \ldots, t+m_i(t)-1$ while $p_{i,t}$ remains posted, as firm j does or does not respond (Figure 8)
    $\widehat{Q}_{i,t} \leftarrow$ accumulated profit + discounted continuation value ▷ two-turn case: Eq. (3)
end if
$Q_{i,t+1}(s_{i,t}, a_{i,t}) \leftarrow (1 - \alpha_t)\, Q_{i,t}(s_{i,t}, a_{i,t}) + \alpha_t\, \widehat{Q}_{i,t}$

The timing structure determines what the target $\widehat{Q}_{i,t}$ credits. In the simultaneous benchmark, the target is one period of profit plus continuation value:

$$\widehat{Q}^{sim}_{i,t} = \pi_i(p_{i,t}, p_{j,t}) + \beta \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+1}, a').$$

Under response-window timing, a chosen price remains active until the firm next receives a pricing opportunity. The target therefore credits the realized profit path while that price is posted, including profit earned after any rival response, plus continuation value from the resulting state. In the usual two-turn case,

$$\widehat{Q}^{rw}_{i,t} = \pi_i(p_{i,t}, p_{j,t}) + \beta \pi_i(p_{i,t+1}, p_{j,t+1}) + \beta^2 \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+2}, a'). \tag{3}$$

Appendix Section C.1 gives the exploration schedule, the exact response-opportunity rule, and the general target for cases in which a response window is not used.

4.2 Simulation Design

Main simulation treatments. The main simulations vary the two forces highlighted above: demand advantage and response reliability. Firm D is fixed as the same firm across all four cells: the treatments turn its demand advantage and response-reliability advantage on or off, while firm S is its rival. Table 4 reports the resulting 2×2 design. In all four cells, both firms use Q-learning and prices follow the response-window timing described above. Only in the bottom-right cell does firm D hold both advantages; this is the dominant-followership treatment referenced throughout the results below. These treatments are the primary comparison for the price results in Table 6.

Table 4. Simulation Treatments

	Response timing	
Demand environment	Symmetric response	Asymmetric response
Symmetric demand	$\lambda = 0.5,\ (\rho_D, \rho_S) = (0.5, 0.5)$	$\lambda = 0.5,\ (\rho_D, \rho_S) = (1, 0)$
Asymmetric demand	$\lambda = 0.7,\ (\rho_D, \rho_S) = (0.5, 0.5)$	$\boxed{\lambda = 0.7,\ (\rho_D, \rho_S) = (1, 0)}$

Notes: In the symmetric-demand row, D and S split demand equally at price parity. In the asymmetric-demand row, D receives share $\lambda = 0.7$ at price parity. In the symmetric-response column, both firms have the same response probability. In the asymmetric-response column, D is the reliable responder.

A separate simultaneous-updating benchmark shuts down the response channel entirely and compares $\lambda = 0.5$ with $\lambda = 0.7$, holding the price grid and learning rule fixed. This benchmark isolates the demand effect before introducing any response channel.

One-learner decomposition under dominant followership. The two-learner simulations show what happens when both firms adapt at the same time, where both demand asymmetry and response-reliability asymmetry are present. Because both firms are learning simultaneously, however, this alone cannot reveal whose learning is necessary for the resulting price increase. To identify this, I hold the dominant-followership environment fixed, at $\lambda = 0.7$ and $(\rho_D, \rho_S) = (1, 0)$, and let one firm at a time follow a myopic best response instead of Q-learning. The myopic firm chooses the price that maximizes current-period profit against the rival's last posted price, without updating continuation values or internalizing how its choice affects future states.[12] Holding one side fixed also removes the moving-target problem created when two algorithms continually adapt to each other, giving the learning firm a stationary environment with a well-defined solution. Table 5 summarizes the three resulting cases; Table 7 below reports which one raises the market price.

Table 5. One-Learner Decomposition: Design (Dominant-Followership Environment)

	Firm S	
Firm D	Q-learning	Myopic best response
Q-learning	$\boxed{\text{Both firms learn}}$	Dominant firm only learns
Myopic best response	Rival only learns	*(excluded: no learning to identify)*

Notes: Each cell fixes one pricing rule per firm. The excluded cell is omitted because, with neither firm learning, there is no learning process left to decompose. The three remaining cells isolate whose learning drives the market price under dominant followership; the boxed cell is the two-learner baseline against which the other two cases are compared.

Outcome measurement. The main outcome is the transaction price, $p_t^{mkt} = \min\{p_{D,t}, p_{S,t}\}$, the price paid by consumers under the baseline demand allocation. I also report firm-level prices and profits to show how learning changes the distribution of gains. Unless otherwise stated, outcomes are averaged over the final 5 percent of periods across simulation runs.

4.3 When Do Learned Prices Rise?

Demand advantage alone does not raise learned prices. Only under dominant followership, when the demand-advantaged firm is also the one that responds reliably, does the price climb substantially above what firms learn without a response channel.

Simultaneous no-response benchmark. The simultaneous-updating benchmark removes the response channel and isolates the demand effect. The learned transaction price is 0.339 under symmetric demand and 0.274 under asymmetric demand: without a response channel, demand asymmetry lowers rather than raises learned prices, even in repeated learning. Appendix Figure A.4 shows the corresponding price trajectories.

[12] Ties between matching and undercutting are broken toward matching, consistent with the weak matching condition in equation (1).

Table 6. LEARNED TRANSACTION PRICES: DEMAND ADVANTAGE AND RESPONSE TIMING INTERACT

	Response timing	
	Symmetric response	Asymmetric response
Symmetric demand	0.388	0.533
Asymmetric demand	0.433	$\boxed{0.660}$

Notes: The table reports average transaction prices over the final 5 percent of periods in the two-learner Q-learning simulations. The transaction price is $p_t^{mkt} = \min\{p_{D,t}, p_{S,t}\}$.

Prices rise only when both asymmetries combine. Table 6 shows that the price increase requires the two asymmetries to coincide. Demand asymmetry alone raises the market price only modestly, from 0.388 to 0.433. Asymmetric response timing alone has a larger effect, raising the price to 0.533. The largest increase occurs under dominant followership, when the demand-advantaged firm is also the reliable responder: the learned market price reaches 0.660, mirroring the mechanism in Section 3. Appendix C.3 shows that, under dominant followership, play concentrates increasingly in the high-price region over the course of training, even though the exact price pair continues to adjust.

Matching becomes far more likely under dominant followership. The key learning object is whether a higher-price attempt is rewarded or punished by the rival's response. Figure 9 measures this by asking whether firm D, which holds the demand and response-reliability advantages when those treatments are turned on, matches after firm S posts a price above D's inherited posted price.

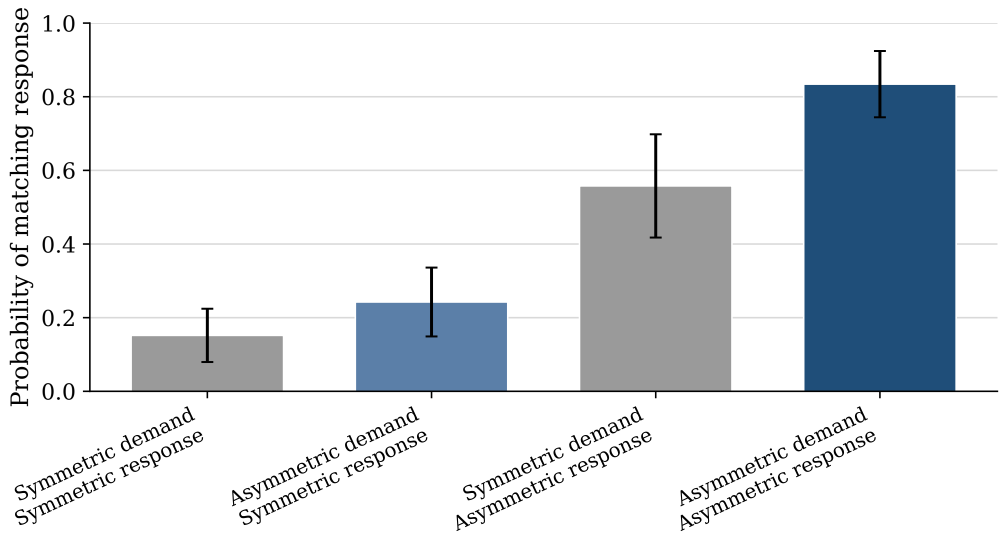

Figure 9. Firm D's Response to Rival Higher Prices.

Notes: Bars report the mean, across simulation runs, of the probability that firm D matches after firm S posts above D's inherited posted price, using events in the final 5 percent of periods. Whiskers report 95 percent confidence intervals across simulation runs.

The figure shows that matching becomes much more likely when the two asymmetries operate together. Demand asymmetry alone raises the matching probability modestly, from 0.15 to 0.24 under symmetric response. Response timing alone has a larger effect, raising it from 0.15 to 0.56 under symmetric demand. The largest effect occurs when the reliable responder is also demand advantaged: the matching probability reaches 0.83. The dominant firm's response turns the rival's higher-price attempts from a risk into a reward. Appendix Figure A.6 decomposes the nonmatching responses.

Play concentrates in high-price states under dominant followership. Figure 10 shows where the algorithms spend time after learning, pooling posted-price pairs across simulation runs in the final part of training.

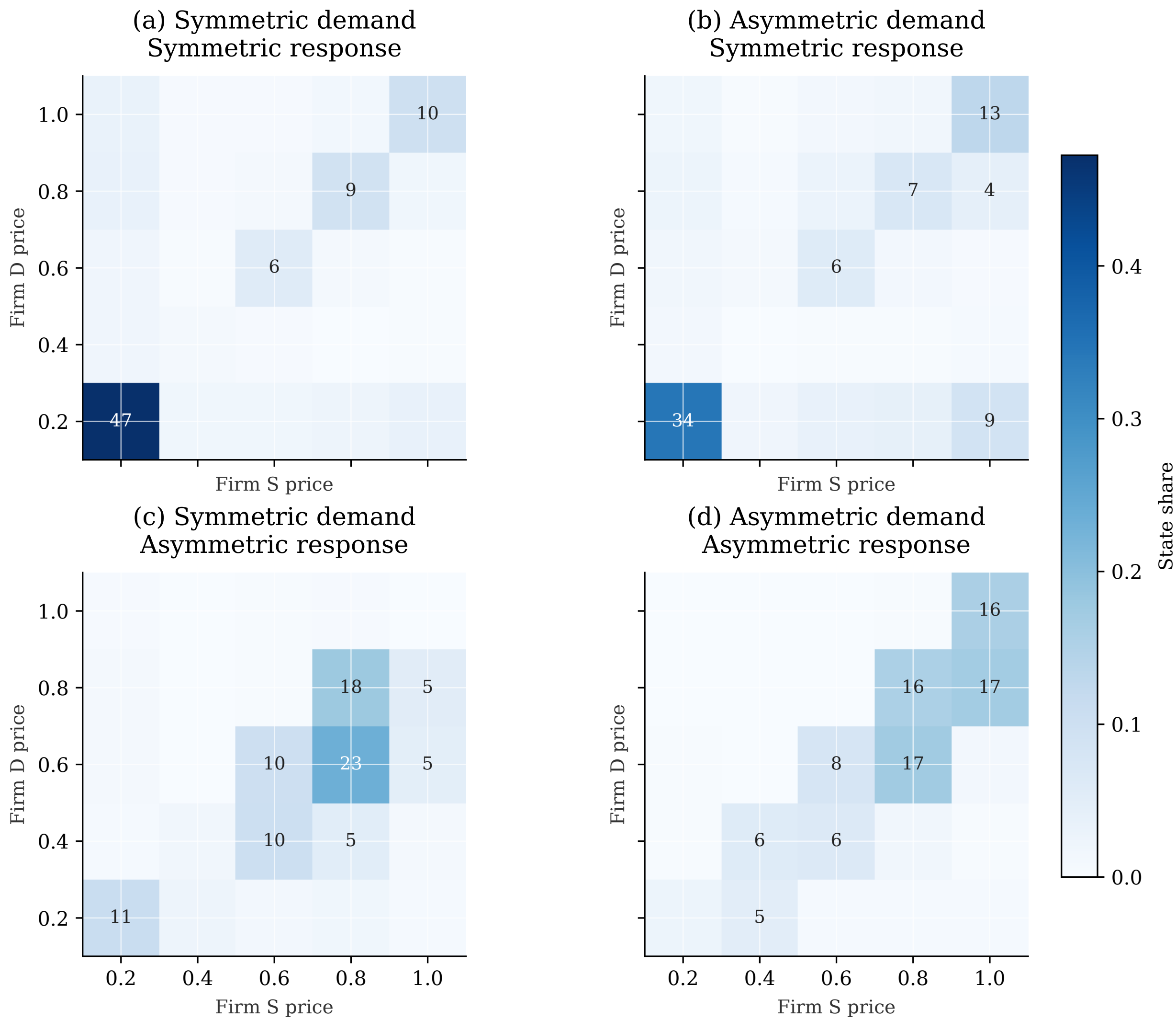

Figure 10. Visited Price States after Learning.

Notes: The figure pools posted-price pairs across all simulation runs in the final 5 percent of training. The vertical axis is firm D's posted price and the horizontal axis is firm S's posted price. Cell labels report percentage shares and are shown for cells with shares of at least 4 percent.

The heatmaps show that the high average prices in Table 6 are not driven by isolated price spikes. In the symmetric-demand, symmetric-response benchmark, nearly half of late-period observations remain at the lowest price pair. The two intermediate treatments show the same ranking as Table 6: the isolated demand effect leaves most mass concentrated near the lowest price pair, while the isolated response-timing effect shifts mass into mid-to-high price cells. Under dominant followership, both firms post high prices in most late-period observations. This matches the response feedback documented above: when higher-price attempts are repeatedly matched, late-period play settles into high-price states.

4.4 Whose Learning Sustains Higher Prices?

Section 3 shows that the higher price is sustained by the dominant firm's willingness to match rather than undercut. One might therefore expect the dominant firm's algorithm to be the source of the price increase in the learning simulations. The one-learner benchmarks show the opposite. Prices rise when only the rival learns, not when only the dominant firm learns.

The asymmetry comes from how much each firm must discover for dominant followership to emerge through learning. The dominant firm observes the rival's higher price before choosing whether to match. The rival must raise price first, absorb the immediate market response, and learn whether the dominant firm later makes that higher price profitable. This mirrors Proposition 2's threshold for costly initiation: the rival must weight future profit heavily enough to justify the demand it sacrifices today; Q-learning offers one way to discover this weighting from repeated market feedback rather than from assumed rationality. The one-learner benchmarks isolate each firm's learning problem to identify which one is responsible for the price increase.

Table 7. Learned Prices by Learner Identity (Dominant-Followership Environment)

	Firm S	
Firm D	Q-learning	Myopic best response
Q-learning	$(0.667, 0.776)$	$(0.248, 0.232)$
Myopic best response	$\boxed{(0.592, 0.598)}$	*(excluded)*

Notes: Each cell reports (p_D, p_S), average posted prices over the final 5 percent of periods across 30 simulation runs, under the dominant-followership treatment ($\lambda = 0.7$, $\rho_D = 1$, $\rho_S = 0$). The market price is the period average of $\min\{p_{D,t}, p_{S,t}\}$, which need not equal $\min\{p_D, p_S\}$ computed from the cell averages shown here, since the lower-priced firm can vary across periods. When only one firm uses Q-learning, the other follows a myopic current-period best response; the excluded cell is omitted since, with neither learning, there is no learning process to decompose.

Table 7 shows that learning matters mainly on the initiation side. When only the dominant firm learns, the market price remains low, at 0.225. When only the rival learns, the market price rises to 0.589, close to the two-learner outcome of 0.660. The algorithm that drives the higher price is therefore not the dominant firm's algorithm, but the rival's.

4.5 Who Captures the Gains?

The rival's algorithm drives the price increase, but it does not capture most of the gains. This is the final asymmetry in the simulation results. Starting from the benchmark in which the dominant firm uses Q-learning and the rival follows a myopic best response, I let the rival switch to Q-learning. Figure 11 shows that this raises the market price by 0.435, from 0.225 to 0.660. Both firms' profits rise, but the gains are highly asymmetric: the dominant firm's profit increases by 0.397, compared with 0.039 for the rival.

Thus, the algorithm that raises prices and the firm that benefits most are not the same. The rival's learning makes higher prices easier to sustain, while the dominant firm's demand advantage allows it to capture most of the resulting surplus. This is the asymmetry behind the Project Nessie allegation described in the introduction: a dominant firm like Amazon need not run the sophisticated algorithm itself to exploit higher prices, so long as its rivals' algorithms learn to raise them.

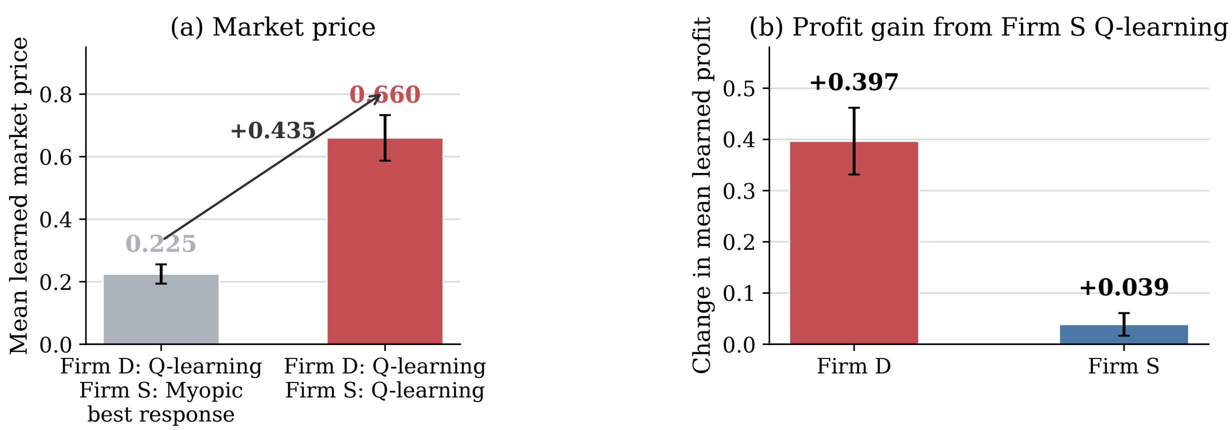

Figure 11. Market-Price and Profit Effects of the Rival's Q-Learning.

Notes: The figure uses the asymmetric-demand, asymmetric-response environment. It compares the benchmark in which the dominant firm uses Q-learning and the rival uses a myopic best response with the two-learner environment in which both firms use Q-learning. Panel (a) reports the change in the market price. Panel (b) reports changes in firm profits. Whiskers report 95 percent confidence intervals across 30 simulation runs. All outcomes are averaged over the final 5 percent of simulation periods.

Appendix C.5 shows that this asymmetric incidence holds at the run level as well: across simulation runs, those in which the rival posts a higher average price than the dominant firm also tend to be the runs in which the dominant firm earns the higher profit premium.

5 Conclusion

Can a dominant firm raise market prices by following rivals rather than leading them? The central lesson of this paper is that it can: dominance can operate through response rather than leadership. A demand-advantaged firm need not be the first mover, or even the firm whose algorithm learns to raise prices. If its response to rival increases is reliable and privately optimal, that response can change what rivals learn to do.

The evidence, model, and simulations identify different parts of the same mechanism. The data show the critical response pattern: Amazon often follows and matches rival price increases. The model explains why following can be privately optimal for a demand-advantaged firm. The simulations show how the same pattern can emerge from market feedback, with the rival's algorithm driving the price increase and the dominant firm capturing most of the gains.

This mechanism should not be evaluated only through price levels, or through whether firms explicitly coordinate: each response can be individually optimal given a firm's own demand position, without any agreement or common pricing rule. Yet once such responses become systematic and learnable, they can change what rival algorithms learn to be profitable, so a routine price response can become part of a feedback system that softens competition. The relevant object to measure is therefore the response system itself: whose price changes become visible first, who responds, how reliably, and whether the responder has a demand advantage. This is the same question raised by the Project Nessie allegation: counterfactual audits should ask whether market outcomes would change if any of these features of the response system were different.

Simultaneous-move models remove this object by construction. Algorithmic price competition among asymmetric firms should therefore be modeled as a sequential learning environment, where the firm whose algorithm learns to raise prices and the firm that captures the resulting gains need not be the same.

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A Empirical Appendix

A.1 Response Robustness and Heterogeneity

Table A.1. Price Adjustment Responses to Rival Price Changes

	Panel A: Any rival move		Panel B: Isolated rival move	
	Price increase	Price decrease	Price increase	Price decrease
Amazon	15.90	8.91	12.94	12.78
	(3.19), $p < .001$	(3.30), $p = .007$	(1.79), $p < .001$	(1.71), $p < .001$
Walmart	0.82	1.93	4.04	1.30
	(2.11), $p = .697$	(1.96), $p = .324$	(1.86), $p = .030$	(2.02), $p = .521$
Target	3.41	2.37	4.42	4.07
	(1.42), $p = .017$	(1.66), $p = .154$	(1.34), $p = .001$	(1.22), $p = .001$
Best Buy	-0.83	-0.94	1.93	1.47
	(1.29), $p = .521$	(1.53), $p = .539$	(1.34), $p = .149$	(1.21), $p = .224$
Home Depot	4.25	4.37	6.12	8.30
	(5.18), $p = .412$	(4.40), $p = .321$	(2.56), $p = .017$	(3.22), $p = .010$
Product FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Clustered SE	Product	Product	Product	Product
Observations	597,309		597,334	

Notes: The unit of observation is platform-product-time. The responding platform is required not to change price at t. Panel A uses any rival price move for the same product at t. Panel B restricts to isolated rival price moves, in which no platform raised price for the product in the prior 48 hours, exactly one identified rival platform moves at t, and the responding platform's own price is unchanged at that moment. Coefficients are percentage-point effects. Standard errors, clustered by product, are reported in parentheses, followed by the exact two-sided p-value.

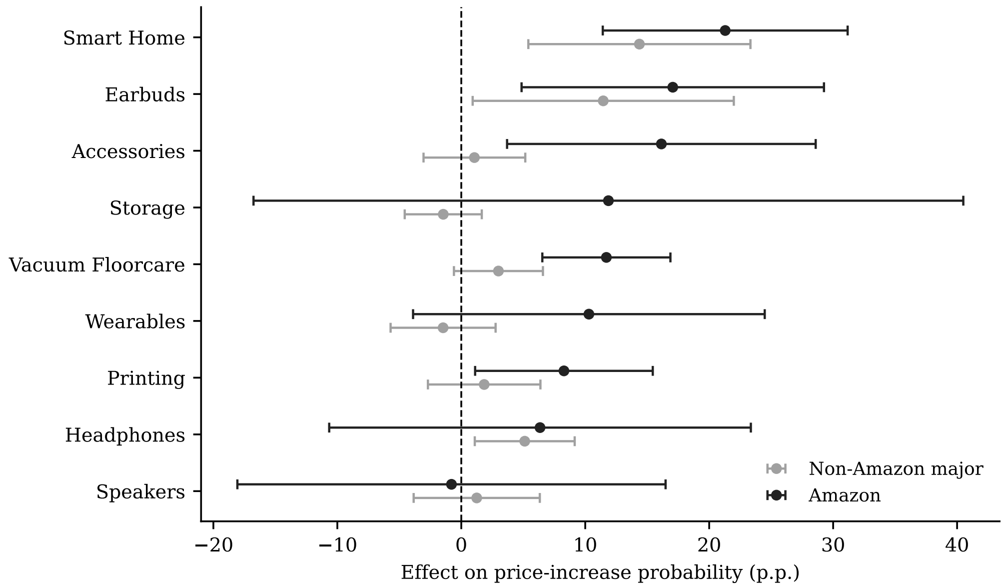

Figure A.1. Amazon Response to Isolated Rival Increases by Product Category.

Notes: The figure reports category-specific versions of the isolated-rival-increase response specification. Black markers report the Amazon coefficient; gray markers report the pooled coefficient for non-Amazon major platforms. Horizontal bars report 95 percent confidence intervals. Each specification includes product, week, day-of-week, and platform fixed effects, with standard errors clustered by product. The figure includes categories with at least 20 Amazon-facing isolated rival increase events and at least five product clusters.

Table A.2. Amazon Response to Isolated Rival Increases by Product Demand Tier

	Low demand (50–2K)	Middle demand (2K–8K)	High demand (9K–100K)
Amazon	8.28	10.94	15.50
	(5.02), $p = .099$	(2.75), $p < .001$	(2.55), $p < .001$
Non-Amazon major platforms	2.35	2.71	4.36
	(3.00), $p = .435$	(1.28), $p = .034$	(1.27), $p < .001$
Products	59	62	67
Observations	144,795	190,916	255,024

Notes: Demand tiers are terciles of each product's median Amazon "Bought in Past Month" demand proxy, computed over the full matched-product panel; the ranges shown are the approximate floor-coded bands from Amazon's own display and are not exact boundaries. Entries are percentage-point effects on the probability of increasing price within 48 hours of an isolated rival price increase, estimated jointly for Amazon and the pooled non-Amazon major platforms (Walmart, Target, Best Buy, and Home Depot) within each tier. Standard errors, clustered by product, are reported in parentheses, followed by the exact two-sided p-value. All specifications include product, week, day-of-week, and platform fixed effects.

A.2 Empirical State Dependence in Prices

The learning simulations use a parsimonious state: the platform's own lagged price and the lagged lowest rival price. Table A.3 supports this choice. Own prices are highly persistent, and the lagged lowest rival price helps explain the focal platform's current price even after conditioning on its own lagged price. The table also shows that a platform's position relative to the lowest rival price predicts the direction of its next adjustment: platforms priced above the rival benchmark tend to adjust downward, while platforms priced below it tend to adjust upward.

Table A.3. State-Space Motivation: Own and Rival Price States

	(1)	(2)	(3)
	$\log p_{ip,t}$	$\log p_{ip,t}$	$\Delta \log p_{ip,t}$
$\log p_{ip,t-1}$	0.9796***	0.9761***	
	(0.0046)	(0.0058)	
$\log p^{\min}_{-i,p,t-1}$		0.0080***	
		(0.0030)	
$\Delta \log p^{\min}_{-i,p,t-1}$			0.0105
			(0.0070)
$Gap_{ip,t-1}$			-0.0159***
			(0.0045)
Product FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Clustered SE	Product	Product	Product
Observations	878,992	878,992	849,328
R^2	0.999	0.999	0.008

Notes: The unit of observation is firm-product-time. The lag $t-1$ is the previous observed state for the same firm-product pair, restricted to observations between 1 and 6 hours apart. $p^{\min}_{-i,p,t}$ is the lowest rival price in the same product market, Δ denotes log price changes, and $Gap_{ip,t-1} = \log p_{ip,t-1} - \log p^{\min}_{-i,p,t-1}$. The sample restricts Amazon and Walmart to first-party listings where seller identity is observed. Standard errors are clustered by product. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Theoretical Appendix

B.1 Comparative Statics for Sequential Matched Prices

Figure A.2 plots the highest sustainable matched price as a function of demand asymmetry and the price-grid increment.

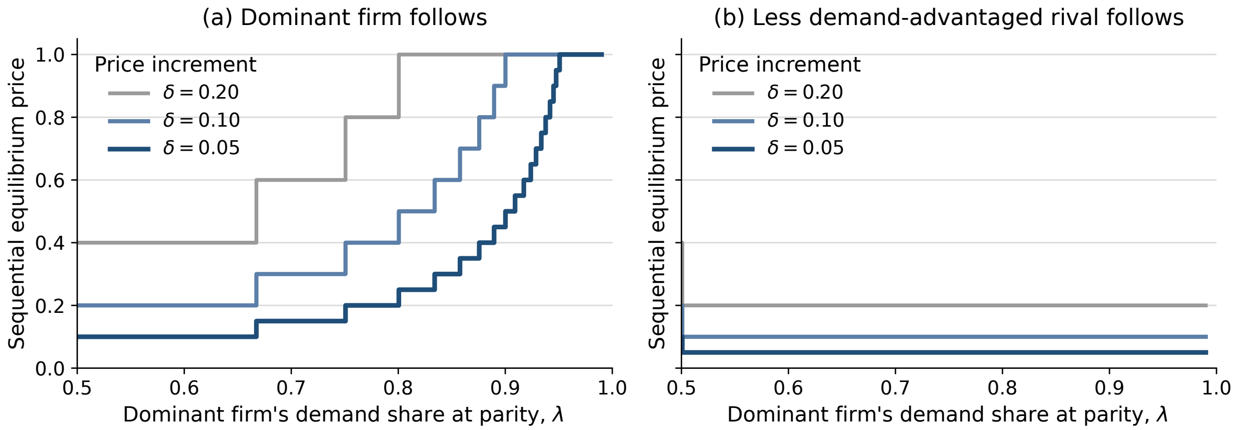

Figure A.2. Sustainable Matched Prices under Sequential Pricing.

Notes: Panel (a) shows the case in which the dominant firm follows, while Panel (b) shows the case in which the rival follows. Prices are the highest sustainable matched prices under the relevant follower constraint. Different lines correspond to different price-grid increments δ; smaller δ gives a finer price grid, so equilibrium prices adjust in smaller steps.

B.2 Behavioral Separation and Role Preference

Proposition 1 has two implications. First, there is a price region in which the dominant follower prefers matching while the rival follower prefers undercutting, so the same higher price can be sustained only when the dominant firm follows. Second, among matched sequential outcomes, both firms can prefer the role assignment in which the demand-advantaged firm follows.

Corollary 2 (Behavioral separation under demand asymmetry)**.** *Suppose $\lambda > 1/2$. For any first-mover price $p \in \mathcal{P}$ satisfying*

$$\frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda},$$

the dominant firm matches as the follower, while the rival undercuts.

Let $\Delta_i^U(p)$ denote firm i's net gain, as follower, from undercutting a first-mover price p, relative to matching. For the dominant firm,

$$\Delta_D^U(p) = (1-\lambda)(p-\delta) - \lambda\delta = (1-\lambda)p - \delta.$$

For the rival,

$$\Delta_S^U(p) = \lambda(p-\delta) - (1-\lambda)\delta = \lambda p - \delta.$$

The incentives have opposite signs exactly when

$$\Delta_D^U(p) < 0 < \Delta_S^U(p) \quad \Longleftrightarrow \quad \frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda}.$$

Corollary 3 (Role preference among matched outcomes)**.** *Among matched sequential outcomes, both firms weakly prefer $S \to D$ to $D \to S$.*

Let $S \to D$ denote the outcome in which the rival moves first and the dominant firm follows, and let $D \to S$ denote the reversed role assignment. Under $S \to D$, the matched price is p_D^{follow}; under $D \to S$, it is p_S^{follow}. At any matched price p, payoffs are

$$\Pi_D(p) = \lambda p, \qquad \Pi_S(p) = (1-\lambda)p.$$

Thus, $\Pi_i^{S\to D} = \Pi_i(p_D^{follow})$ and $\Pi_i^{D\to S} = \Pi_i(p_S^{follow})$. Since Proposition 1 gives $p_D^{follow} \geq p_S^{follow}$, both firms earn weakly higher payoffs under $S \to D$ among matched outcomes.

B.3 Proof of Proposition 2 and Corollaries 4, 1

Corollary 4 (A verified equilibrium)**.** *On the normalized five-point grid, for $\lambda \in (2/3, 3/4)$, $p^M = 3\delta$ and $\bar{\beta} = 1/2$. In this region, for $\beta > 1/2$, δ is also the dominant firm's own optimal initial price, so the unique equilibrium path is*

$$\delta \to 3\delta \to 3\delta.$$

The game is solved by backward induction.

The dominant firm's final response. The dominant firm's final response solves

$$a_3^*(a_2) \in \arg\max_{a_3 \in \mathcal{P}} \pi_D(a_3, a_2).$$

The discount factor does not affect this final choice, because only the second market payoff remains. Given a rival price p, matching yields λp, while undercutting by one grid step yields $p - \delta$. Thus the dominant firm weakly prefers to match p if and only if

$$\lambda p \geq p - \delta,$$

or $p \leq \delta/(1-\lambda)$. Let

$$p^M \equiv \left\lfloor \frac{\delta}{1-\lambda} \right\rfloor_{\mathcal{P}}$$

denote the highest grid price the rival can induce the dominant firm to match.

The rival's response threshold. Suppose the dominant firm initially posts δ, and suppose $p^M > \delta$. The rival's response solves

$$a_2^*(a_1) \in \arg\max_{a_2 \in \mathcal{P}} \left\{ \pi_S(a_1, a_2) + \beta \pi_S\big(a_3^*(a_2), a_2\big) \right\}.$$

If the rival ties at δ, the lowest grid price, its payoff is $(1+\beta)(1-\lambda)\delta$. If the rival instead raises price to p^M, it loses demand at the first clearing but is matched by the dominant firm in the final revision, giving payoff $\beta(1-\lambda)p^M$. Lower price increases yield a smaller matched continuation price, while prices above p^M are not matched. The rival therefore prefers raising price to p^M whenever

$$\beta(1-\lambda)p^M > (1+\beta)(1-\lambda)\delta \quad \Longleftrightarrow \quad \beta > \bar{\beta}(\lambda) \equiv \frac{\delta}{p^M - \delta}.$$

Since $p^M = \lfloor \delta/(1-\lambda) \rfloor_{\mathcal{P}}$ is a step function that weakly increases in λ, the threshold $\bar{\beta}(\lambda)$ is weakly decreasing in λ.

Proof of Corollary 1. Suppose instead that the dominant firm's final turn is gated only after a rival price change. If the rival ties at δ, then $a_2 = a_1$, so the dominant firm's final turn is ordinary and ρ_D does not enter. The dominant firm has no reason to move away from the tie, so the rival's payoff from tying is $(1+\beta)(1-\lambda)\delta$. If the rival raises price to p^M, then $a_2 \neq a_1$, so the dominant firm's final turn is a response window. The rival is matched only when that window activates, giving expected payoff $\beta\rho_D(1-\lambda)p^M$. The rival prefers raising price to p^M whenever

$$\beta\rho_D(1-\lambda)p^M > (1+\beta)(1-\lambda)\delta \quad \Longleftrightarrow \quad \beta > \bar{\beta}(\lambda, \rho_D) \equiv \frac{\delta}{\rho_D\, p^M - \delta},$$

which reduces to $\bar{\beta}(\lambda)$ when $\rho_D = 1$. If instead $\rho_D p^M \leq \delta$, the left side is non-positive for any $\beta > 0$ while the right side is strictly positive, so the inequality fails for every β; the division above is invalid in this case, and no discount factor rationalizes initiation.

Verifying the equilibrium for a specific region. The threshold $\bar{\beta}(\lambda)$ is conditional on the dominant firm posting δ. It remains to verify that δ is the dominant firm's own best initial choice. Consider $\lambda \in (2/3, 3/4)$, so that $\delta/(1-\lambda)$ lies between 3δ and 4δ and $p^M = 3\delta$. Then $\bar{\beta} = 1/2$, so for $\beta > 1/2$ the rival's best response to an initial dominant price of δ is

$$a_2^*(\delta) = 3\delta.$$

The rival sacrifices current demand to create a higher matched continuation state.

Applying the same response problem to the other possible initial dominant prices, for $\beta > 1/2$ in this region, gives

$$a_2^*(2\delta) = \delta, \quad a_2^*(3\delta) = 2\delta, \quad a_2^*(4\delta) = 3\delta, \quad a_2^*(5\delta) = 4\delta.$$

Finally, the dominant firm's initial choice solves

$$a_1^* \in \arg\max_{a_1 \in \mathcal{P}} \left\{ \pi_D\big(a_1, a_2^*(a_1)\big) + \beta \pi_D\big(a_3^*(a_2^*(a_1)), a_2^*(a_1)\big) \right\}.$$

Given the response policy above, choosing δ gives the dominant firm current market demand and induces the rival to choose 3δ, which the dominant firm then matches. The resulting payoff, $\delta + 3\beta\lambda\delta$, strictly exceeds the payoff from every other initial price in this region. Hence, the unique equilibrium path is

$$\delta \to 3\delta \to 3\delta.$$

B.4 Reverse Timing in the Two-Clearing Game

Consider instead the order $S \to D \to M \to S \to M$, so that the rival receives the final revision opportunity. The rival's final response is the same follower problem behind p_S^{follow} in Proposition 1: it matches p if and only if $p \leq \delta/\lambda$. Since $\lambda > 1/2$, $\delta/\lambda < 2\delta$, so the rival's matching set never extends above the lowest grid price:

$$a_3^*(p) = \begin{cases} \delta, & p = \delta, \\ p - \delta, & p > \delta. \end{cases}$$

Consequently, whatever price the dominant firm is induced to match at the first clearing, the rival undoes it at the final revision whenever that price exceeds δ. For example, at $\lambda \in (2/3, 3/4)$ and $\beta < 3(1-\lambda)$, backward induction gives the equilibrium path

$$3\delta \to 3\delta \to 2\delta :$$

the rival initiates and the dominant firm matches at the first clearing, exactly as in the forward case, but the rival then undercuts its own initiated price at the final revision.

B.5 Proof of Proposition 3

Fix the candidate policy in Section 3.4. The proof verifies this policy by solving its continuation values and checking one-period deviations. Once the policy is fixed, each state has a value: current profit from the prescribed price choice plus the discounted value of the next state generated by the revision process. Thus, for firm i, the value of state (p_D, p_S, m, c) satisfies

$$V_i(p_D, p_S, m, c) = \pi_i\big(p'_D(\sigma_m), p'_S(\sigma_m)\big) + \beta\mathbb{E}\left[V_i(p'_D(\sigma_m), p'_S(\sigma_m), m', c')\right],$$

where only the active firm's price can change before the market clears, m' is the next scheduled mover, and c' records whether the active firm changed price. The expectation is over whether a response window activates. These Bellman equations are linear because the policy is fixed. I then check one-period deviations. Since there are only two prices, each deviation replaces the active firm's prescribed price with the other price and compares the resulting payoff with the value of following the policy.

Let $x = L/H$. Solving the Bellman system and checking one-period deviations gives four sufficient conditions. Two admit an exact closed form for every ρ_S:

$$x \leq \lambda, \qquad x \leq \beta.$$

The condition $x \leq \lambda$ is the dominant firm's matching condition: at (L, H), matching gives λH, while undercutting gives L. The condition $x \leq \beta$ is the rival's initiation condition from (L, L): the rival must care enough about the future matched price to give up current demand.

The remaining two conditions rule out, respectively, the dominant firm self-initiating H from (L, L) and the rival undercutting once the market reaches (H, H). Both depend on the continuation value of the off-path state (H, L), which loops back through the full state space rather than resolving in one step, so neither has a simple closed form for general ρ_S. At $\rho_S = 1$ (pure alternation) the loop shortens enough to give

$$\frac{\beta\lambda}{\lambda+\beta} \leq x \qquad \text{and} \qquad x \leq \frac{(1-\lambda)(1+\beta+\beta^2)}{1-\beta\lambda+\beta}.$$

For $\rho_S < 1$, I verify both conditions by solving the Bellman system directly rather than through a closed-form threshold. For example, at $(\lambda, \beta, \rho_D, \rho_S, x) = (0.7, 0.9, 1, 0, 0.65)$, all four deviation gaps are strictly positive. By continuity, the candidate policy is therefore optimal on a nonempty neighborhood of this point. Starting from (L, L), once the rival receives a revision opportunity it chooses H, the dominant firm then matches, and the market converges to (H, H).

The no-undercutting condition becomes easier to satisfy as ρ_S falls. A rival that undercuts must later pass through a response window to return to (H, H), so a less reliable rival expects a longer delay before it can restore the high price. This is why lower ρ_S weakly expands the set of parameters that sustain dominant followership.

B.6 Loyal-Demand Extension

The baseline model isolates the matching-versus-undercutting mechanism by assuming that a firm charging above its rival receives no demand. This extension relaxes that assumption by allowing each firm to retain loyal consumers even when it charges more than its rival. With loyal demand, the dominant follower can monetize demand advantage in two ways: it can match the rival's price, or it can remain above the rival and serve only loyal consumers. Matching must therefore beat not only undercutting, but also staying expensive.

Proposition 4 (Dominant followership with loyal demand)**.** *Fix a leader price $p = k\delta$, with $k \in \{2, \ldots, M-1\}$. In the loyal-switcher environment, there exists a nonempty open set of primitives (α, β, λ), with*

$$0 < \beta < \alpha, \qquad \alpha + \beta < 1, \qquad \lambda \in [1/2, 1),$$

such that the dominant follower's unique best response is to match p, while the rival's unique best response is to undercut p by one grid step. Thus, the dominant-followership response pattern can arise even when both firms retain positive demand above the leader's price.

Figure A.3 shows that this can hold on a nonempty parameter region. Loyal demand narrows the mechanism, but does not eliminate it: when the dominant firm still gains enough from matching and the rival remains sufficiently dependent on switchers, the behavioral separation behind dominant followership survives.

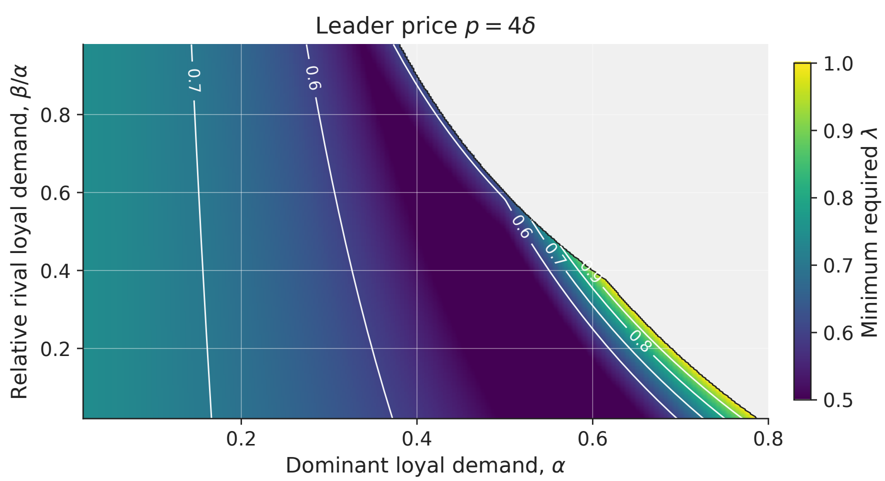

Figure A.3. Dominant Followership with Loyal Demand.

Notes: The figure fixes $M = 5$ and leader price $p = 4\delta$. It plots the minimum λ required for the dominant follower to match while the rival undercuts. Colored regions satisfy $\bar{\lambda}(\alpha, \beta, 4, 5) < 1$; gray regions admit no feasible $\lambda < 1$.

Proof of Proposition 4. Fix a leader price $p = k\delta$, with $k \in \{2, \ldots, M-1\}$. Total demand is normalized to one. A mass α is loyal to the dominant firm, a mass β is loyal to the rival, and the remaining mass

$$s = 1 - \alpha - \beta$$

consists of switchers, with $0 < \beta < \alpha$ and $\alpha + \beta < 1$. Loyal consumers buy from their preferred firm regardless of relative prices. Switchers buy from the lower-price firm, and at equal prices the dominant firm receives share $\lambda \in [1/2, 1]$ of switchers. Demand for the dominant firm is therefore

$$q_D(p_D, p_S) = \begin{cases} \alpha + s, & p_D < p_S, \\ \alpha + \lambda s, & p_D = p_S, \\ \alpha, & p_D > p_S, \end{cases}$$

while demand for the rival is

$$q_S(p_D, p_S) = \begin{cases} \beta, & p_D < p_S, \\ \beta + (1-\lambda)s, & p_D = p_S, \\ \beta + s, & p_D > p_S. \end{cases}$$

The proof first reduces the follower's response problem to three relevant actions, then derives the response inequalities, and finally shows that the feasible set is nonempty.

Step 1: Relevant follower actions. For either follower, matching gives demand at price parity. Among prices below the leader, $p - \delta$ is optimal because any lower price receives the same demand at a lower margin. Among prices above the leader, $M\delta$ is optimal because the firm receives only loyal demand and profit is increasing in price. Thus, it is enough to compare three actions: match, undercut by one grid step, and choose the loyal-only high-price option.

For the dominant follower, the three payoffs are

$$\pi_D^{match} = k\delta[\alpha + \lambda s], \qquad \pi_D^{under} = (k-1)\delta(1-\beta), \qquad \pi_D^{high} = M\delta\alpha.$$

For the rival, they are

$$\pi_S^{match} = k\delta[\beta + (1-\lambda)s], \qquad \pi_S^{under} = (k-1)\delta(1-\alpha), \qquad \pi_S^{high} = M\delta\beta.$$

Step 2: Response inequalities. The dominant follower matches if matching beats both deviations:

$$\pi_D^{match} > \max\{\pi_D^{under}, \pi_D^{high}\}.$$

Substituting the payoffs gives

$$\lambda > \max\left\{\frac{(k-1)(1-\beta)-k\alpha}{ks}, \frac{\alpha(M-k)}{ks}\right\}.$$

The rival undercuts if undercutting beats both alternatives:

$$\pi_S^{under} > \max\{\pi_S^{match}, \pi_S^{high}\},$$

or equivalently,

$$\lambda > \frac{1-\alpha}{ks}$$

and

$$(k-1)(1-\alpha) > M\beta.$$

The first inequality makes undercutting better than matching; the second makes undercutting better than serving only loyal consumers at the high price.

Step 3: Feasibility. Combining the λ-restrictions gives the cutoff

$$\bar{\lambda}(\alpha,\beta,k,M) \equiv \max\left\{\frac{1}{2}, \frac{(k-1)(1-\beta)-k\alpha}{ks}, \frac{\alpha(M-k)}{ks}, \frac{1-\alpha}{ks}\right\},$$

where $s = 1-\alpha-\beta$. The dominant-followership response pattern holds whenever

$$\lambda > \bar{\lambda}(\alpha,\beta,k,M)$$

and

$$(k-1)(1-\alpha) > M\beta.$$

For fixed (α,β,k,M), an admissible $\lambda \in [1/2,1)$ therefore exists if $\bar{\lambda}(\alpha,\beta,k,M) < 1$ and the additional condition holds.

These conditions are not vacuous. For any $\alpha \in (0, k/M)$, choosing $\beta > 0$ sufficiently small makes $\bar{\lambda}(\alpha, \beta, k, M) < 1$ and satisfies $(k-1)(1-\alpha) > M\beta$. Hence there exists $\lambda \in (\bar{\lambda}(\alpha, \beta, k, M), 1)$. Since the inequalities are strict, the response pattern holds on an open set of primitives.

C Simulation Appendix

C.1 Q-Learning Implementation Details

This appendix collects the exact specification of the Q-learning simulations summarized in Section 4.

Response opportunities. Formally, suppose firm i is active in period t, so firm j is scheduled next. Let ρ_j denote the probability that firm j can revise on its next scheduled turn after firm i changes price. Draw $u_{t+1} \sim U[0,1]$ and define

$$R_{j,t+1} = \mathbf{1}\{a_{i,t} \neq p_{i,t-1}\}\mathbf{1}\{u_{t+1} \leq \rho_j\}.$$

This implements the rule illustrated in Figure 8: $p_{j,t+1} = a_{j,t+1}$ and $p_{i,t+1} = p_{i,t}$ when $R_{j,t+1} = 1$, and prices remain unchanged otherwise. The main simulations vary ρ_D and ρ_S to distinguish symmetric from asymmetric response timing.

Exploration. The ϵ_t-greedy exploration rule is given in Algorithm 1, steps 2–7.

General learning target. Section 4 gives the Q-learning update rule and the simultaneous and two-turn posted-price targets. More generally, a scheduled response window may not be used, in which case the market still clears at unchanged posted prices and the firm earns another realized profit before its next pricing decision. Letting $m_i(t)$ denote the number of market clearings between firm i's choice at t and that next pricing decision, the implemented target is

$$\widehat{Q}^{rw}_{i,t}(s_{i,t}, a_{i,t}) = \sum_{k=0}^{m_i(t)-1} \beta^k \pi_i(p_{i,t+k}, p_{j,t+k}) + \beta^{m_i(t)} \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+m_i(t)}, a').$$

This reduces to the two-turn target in Section 4 when $m_i(t) = 2$ and every scheduled window is used.

Parameters. The discount factor is fixed at $\beta = 0.99$. Learning rates and exploration rates follow parallel hyperbolic decay schedules over a simulation horizon of T periods:

$$\alpha_t = \max\left\{\underline{\alpha}, \alpha_0 \frac{\kappa_\alpha}{\kappa_\alpha + t}\right\}, \qquad \epsilon_t = \max\left\{\underline{\epsilon}, \epsilon_0 \frac{\kappa_\epsilon}{\kappa_\epsilon + t}\right\}.$$

In the baseline simulations, $\alpha_0 = \epsilon_0 = 0.1$, $\underline{\alpha} = \underline{\epsilon} = 0.05$, and $\kappa_\alpha = \kappa_\epsilon = 0.1T$, so both learning and exploration decline gradually over the simulation horizon but remain bounded away from zero.

Myopic best response. Formally, the myopic best response used in the one-learner benchmarks (Section 4) is $BR_i(s_{i,t}) \in \arg\max_{p \in \mathcal{P}} \pi_i(p, p_{j,t-1})$, with ties between matching and undercutting broken toward matching, consistent with the weak inequality in equation (1).

C.2 No-Response Learning Benchmark

Figure A.4 plots the transaction-price trajectories for the simultaneous-updating benchmark discussed in the main text. With no response channel, demand asymmetry does not produce higher learned prices.

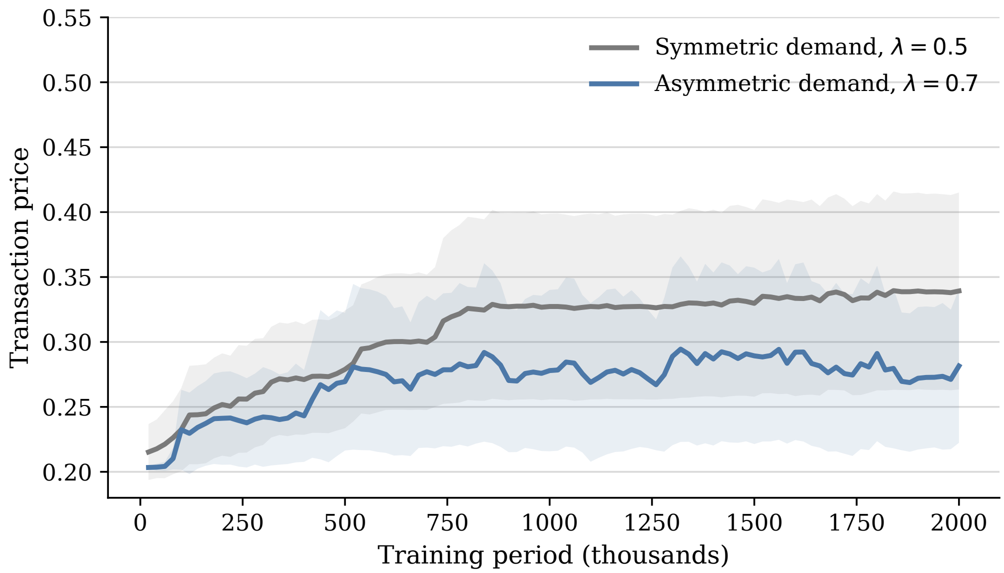

Figure A.4. Transaction Prices under Simultaneous Updating.

Notes: The figure plots average transaction prices in the simultaneous-updating benchmark. Lines average across simulation runs, and shaded regions report 95 percent confidence intervals.

C.3 Late-Period Dynamics in Two-Learner Simulations

The two-learner simulation under dominant followership generates recurrent price adjustment rather than a single fixed price pair. This is the object summarized in Table 6 and Figure 10: late-period play under mutual adaptation.

Learning nevertheless moves play toward a stable region even though the exact price pair does not settle. Figure A.5 shows that the share of periods in which both firms post prices weakly above 0.6 rises during training and is highest when the reliable responder is also demand advantaged: play becomes concentrated in the high-price region even though the specific price pair within it continues to change.

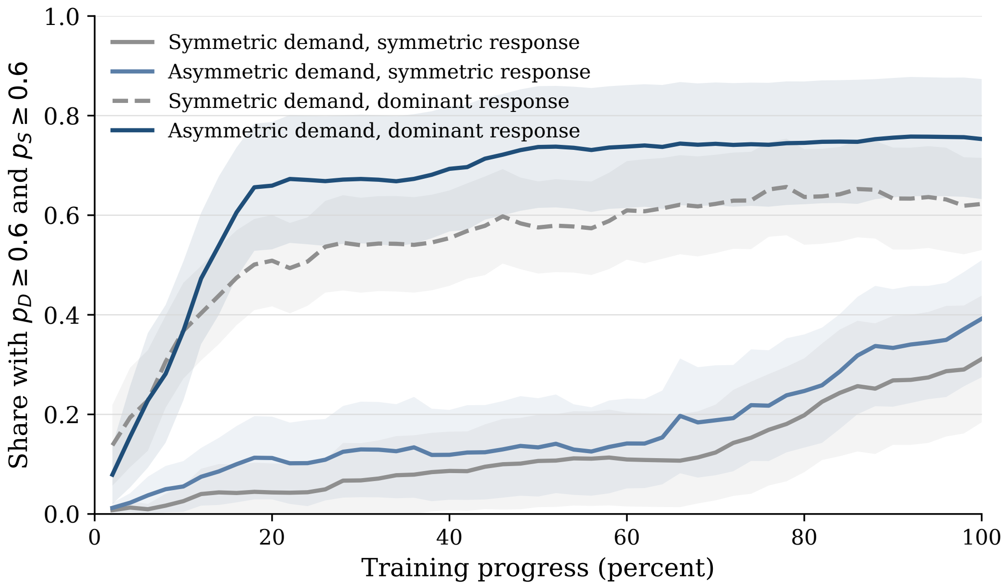

Figure A.5. Learning into High-Price States.

Notes: The figure reports, by training bin, the share of periods in which both firms post prices weakly above 0.6. Lines average across simulation runs; shaded regions report 95 percent confidence intervals across runs.

This interpretation is consistent with prior work on multi-agent Q-learning, where learned play often takes the form of recurrent price dynamics rather than a fixed price profile (Klein 2021; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024). The one-learner decomposition in Table 7 provides the cleaner evidence on the economic channel behind this pattern.

C.4 Response Decomposition

Figure A.6 decomposes firm D's next-period action after the rival posts above its inherited price. The main text focuses on matching because matching reinforces the rival's higher-price move. This figure shows what happens when matching does not occur. In the comparison cells, higher-price moves are often followed by undercutting or no response. When demand advantage and asymmetric response timing are combined, matching accounts for most of firm D's next-period actions and undercutting becomes much less frequent.

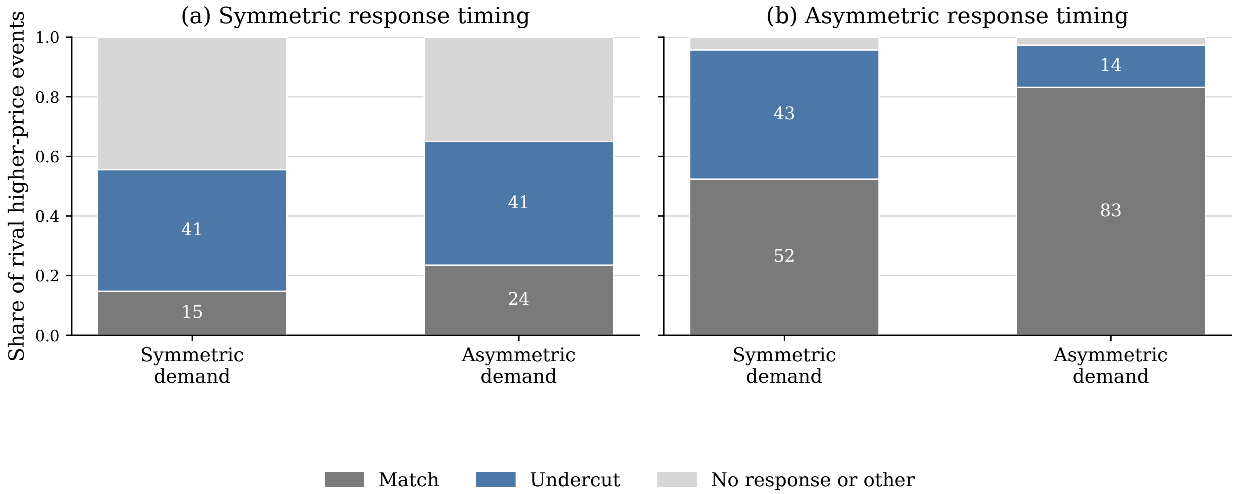

Figure A.6. Response Decomposition after Rival Higher Prices.

Notes: The figure conditions on final-5-percent periods in which firm S posts a price above firm D's inherited posted price. Bars classify firm D's next-period action. A match means firm D sets the same price as firm S; an undercut means it sets a lower price. The remaining category includes cases in which firm D's next posted price is unchanged or is above firm S's price.

C.5 Asymmetric Incidence across Runs

Figure A.7 reports the run-level counterpart to the asymmetric incidence documented in Section 4: it plots, for each simulation run, the rival's posted-price premium against the dominant firm's profit premium. Points in the upper-right region therefore correspond to runs in which the rival posts higher prices on average and the dominant firm earns higher profits. The dominant-followership runs are concentrated in this region, showing that the asymmetric incidence is systematic across runs and not an artifact of period averaging.

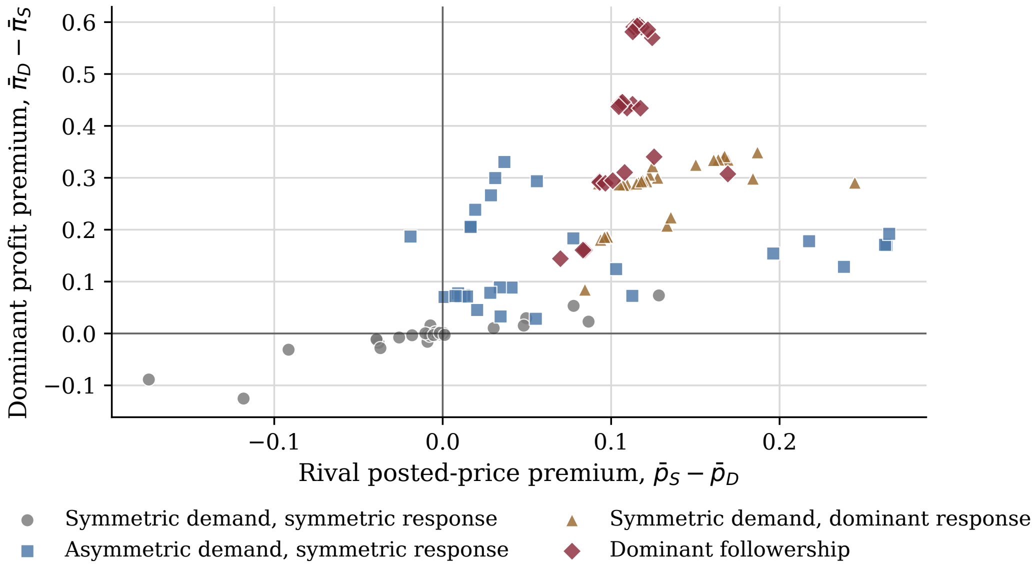

Figure A.7. Run-Level Incidence of Prices and Profits.

Notes: Each point is one simulation run, using averages over the final 5 percent of periods. The horizontal axis reports $\bar{p}_S - \bar{p}_D$, the difference between the rival's average posted price and the dominant firm's average posted price. The vertical axis reports $\bar{\pi}_D - \bar{\pi}_S$, the corresponding dominant-firm profit premium.